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A Co-registered Visible, Thermal-Infrared and Elevation Stack of Mars at ±60°: What It Maps and What It Cannot

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ABSTRACT

Viking MDIM 2.1 visible colour, THEMIS day and night thermal-infrared mosaics and the HRSC/MOLA blended elevation model were co-registered on one grid spanning ±60° latitude, 86.6% of Mars, to map impact craters, fluvial channels and lava flows. The THEMIS mosaics are 8-bit products stretched region by region: dusty and rocky provinces differ by 47.67 DN in Viking red but by −0.19 and +0.02 DN in day and night infrared, and against TES thermal inertia the THEMIS pair keeps only a weak local signal ($|r| \approx 0.1$) while Viking colour tracks inertia planet-wide ($r = -0.36$). Fill depth yields 465,354 closed depressions ≥ 1 km; 63.2% of those ≥ 5 km are catalogued craters, and most craters missed are degraded or breached. Of 969,822 routed channel segments 89.8% lie on slopes under 2°. A support vector machine trained on hand-drawn polygons and scored in whole 15° blocks it never saw reaches a balanced accuracy of 66.7% (73.8% overall), against 61.3% for the nearest training polygon's class; the thermal bands add 1.4 to 2.0 points overall but none by balanced accuracy. Revised to seven classes on a third, rule-made label set, a U-Net scores 87.6% on 237 held-out polygons, 86.6% and 85.4% on two draws of other blocks, and 49.6% on a probability sample. No band separates the geologic map's lava contacts or traced margins at Athabasca Valles much better than chance, so lava flows are taken from that map. Three of four hypotheses are not supported; the fourth is answered by another method.

Keywords: Mars — Planetary surfaces — Remote sensing — Infrared imaging — Impact craters — Lava flows — Fluvial geomorphology — Machine learning

1. INTRODUCTION

Athabasca Valles, a young channel system south of Elysium, was read for decades as an outflow channel carved by water released from the Cerberus Fossae (D. M. Burr et al. 2002; J. B. Plescia 2003). High-resolution imaging later showed the channel floor draped by a single turbulent flood of lava (W. L. Jaeger et al. 2007, 2010), and the USGS geologic map places it in the Late Amazonian volcanic unit *lAv* (K. L. Tanaka et al. 2014). The case makes the question of this project concrete: can a co-registered visible and thermal-infrared mosaic separate lava, water and impact landforms, and how much of the answer comes from the thermal infrared?

Thermal infrared is the natural candidate. Rock and coarse debris hold heat through the night while dust cools quickly, so the diurnal temperature contrast follows thermal inertia, a property of particle size and induration (M. T. Mellon et al. 2000; R. L. Fergason et al. 2006; N. E. Putzig & M. T. Mellon 2007). Visible albedo also separates bright dust from dark basalt (S. W. Ruff & P. R. Christensen 2002), and topography carries the morphology of craters, channels and flow fronts. The global mosaics that hold each of these quantities have been available for years (B. A. Archinal et al. 2003; C. S. Edwards et al. 2011; R. L. Fergason et al. 2018), but they come on three different coordinate frames, and a usable stack of them is not a given.

Automated mapping of Mars has mostly taken one landform at a time. Crater detection has moved from handcrafted image features to convolutional neural networks (D. M. DeLatte et al. 2019), which find craters in lunar elevation models (A. Silburt et al. 2019) and in Mars data (C. Lee 2019); valley networks were mapped across the planet by an algorithm that recognises valleys in topography, its output then edited by eye (W. Luo & T. F. Stepinski 2009). Landform classes have been mapped without labels, from topographic attributes (B. D. Bue & T. F. Stepinski 2006), and with a convolutional network trained on fifteen landforms in Context Camera images (T. Wilhelm et al. 2020). Here the visible, thermal-infrared and elevation mosaics are stacked

first, and the three families are mapped from the one stack.

This paper reports the stack, the landform products made from it and the tests run on them. The goals were (1) one co-registered stack of Viking visible colour, THEMIS day and night infrared and HRSC/MOLA topography on a single grid; (2) maps of the three landform families, first in a type area and then across the whole extent; (3) a test of whether the thermal infrared separates surfaces that visible light cannot; and (4) a score for every product against something it never saw. Four hypotheses were set when the project was proposed: H1, that day infrared shows lava-flow margins under dust that Viking misses; H2, that fluvial channels have shallower gradients than volcanic ones and the two separate on gradient against thermal response; H3, that the multi-band composite gives more stable unsupervised classes than any single input; and H4, that crater rims found from image gradients recover most of the 141 named craters larger than 100 km. Each test below wrote its prediction and verdict rule into its script before it ran.

2. DATA

2.1. *The source mosaics*

Four global or near-global products were used (Table 1). The THEMIS mosaics (P. R. Christensen et al. 2004; C. S. Edwards et al. 2011) are single-band 8-bit display products: the night mosaic's label gives a base of 0 and a multiplier of 1, with no scaling to radiance or temperature, and the two mosaics were stretched independently (2nd and 98th percentiles of 35 and 205 DN by day, 58 and 203 DN by night over Ius Chasma). Calibrated thermal inertia cannot be derived from them. Viking MDIM 2.1 has every one of its 4.25 billion pixels valid, but it is not calibrated albedo either: its images were high-pass filtered at about 50 km to remove regional albedo and normalised so that equal slopes show equal contrast, and the colour product is an older colourised mosaic blended over it (B. A. Archinal et al. 2003; USGS Astrogeology Science Center 2009). The Viking bands therefore carry topographic shading and local albedo under an uncalibrated colour, called Viking colour here. The HRSC/MOLA blend (D. E. Smith et al. 2001; R. Jaumann et al. 2007; R. L. Fergason et al. 2018) is signed 16-bit metres on a geographic grid.

2.2. *Reference data*

Four independent references score the products (R. G. Congalton 1991). The USGS Geologic Map of Mars, SIM 3292 (K. L. Tanaka et al. 2014), supplies 1,311 unit polygons and is the lava-flow reference. The Robbins and Hynek database (S. J. Robbins & B. M. Hynek 2012), 2020 release, holds 385,049 craters ≥ 1 km. TES thermal inertia (P. R. Christensen et al. 2001; N. E. Putzig & M. T. Mellon 2007), dayside and nightside at 20 pixels per degree, is the only calibrated thermal product. The IAU gazetteer (IAU Working Group for Planetary System Nomenclature 2019) names 141 craters larger than 100 km. Every IAU crater larger than 100 km matches a Robbins crater with a median centre offset of 1.0% of its diameter, and the Athabasca Valles floor falls in unit *lAv* of the geologic map, as published.

2.3. *The analysis extent and the type areas*

No global THEMIS night mosaic exists: coverage stops at ±60°. The analysis stack was therefore bounded at ±60°, which removes a third of the rows of a simple cylindrical grid but only 13.4% of the surface: 86.6% of Mars lies within ±60°. Relief inside the band equals global relief (29,754 m, Hellas Planitia to Olympus Mons); the polar layered deposits and ice caps, none of them a target landform, are lost. Two type areas prove methods before they run at ±60°: Ius Chasma (271–286°E, 6–13°S), a canyon with the Louros Valles tributaries, and Athabasca Valles (150–162°E, 4–14°N), the flood lava (Figures 1 and 2).

3. METHODS

All processing ran headless through `arcpy` in ArcGIS Pro 3.7 (Esri 2026), with GDAL (GDAL/OGR contributors 2024) and NumPy (C. R. Harris et al. 2020) for raster arithmetic, on a workstation with an Intel i7-8700K, 64 GB of memory and an NVIDIA RTX 2080 Ti; figures were drawn with Matplotlib (J. D. Hunter 2007) or exported from the project's map layouts. Every product and figure has a script that regenerates it (L. M. Edwards 2026). Figure 3 outlines the chain from the source mosaics to the maps.

3.1. *Co-registration on one grid*

The four sources span three frames: THEMIS on a central meridian of 180° (longitudes 0–360°E), Viking on 0° (−180 to 180°E), and the DEM in geographic degrees. A THEMIS window read at face value lands half a planet away, and elevation derivatives computed on the degree grid carry an undefined vertical factor. A global Composite Bands run on the unharmonised sources was cancelled four times by 13 September (and a fifth time on 24 September); projected first and bounded to the type area, the same composite took 8.8 s.

The common grid is read from the night mosaic itself: equidistant cylindrical, central meridian 180°, sphere

Table 1. The source mosaics, as held

Product	Cell	Size (px)	Frame	Holds
Viking MDIM 2.1 colour	231.5 m	92,160 × 46,080 × 3	cylindrical, CM 0°	visible colour, RGB
THEMIS Day IR v12	100 m	213,390 × 106,696	cylindrical, CM 180°	day brightness, 8-bit
THEMIS Night IR v14	100 m	213,388 × 71,130	cylindrical, CM 180°, ±60°	night brightness, 8-bit
HRSC/MOLA blend DEM v2	0.0033741°	106,694 × 53,347	geographic, degrees	elevation, −8.5 to +21.2 km

Note: THEMIS 6.8–14.9 μm. All four are uncompressed and stored one scanline per block; together about 62 GB. The DEM cell is 199.99999 m on the 3,396,190 m sphere.

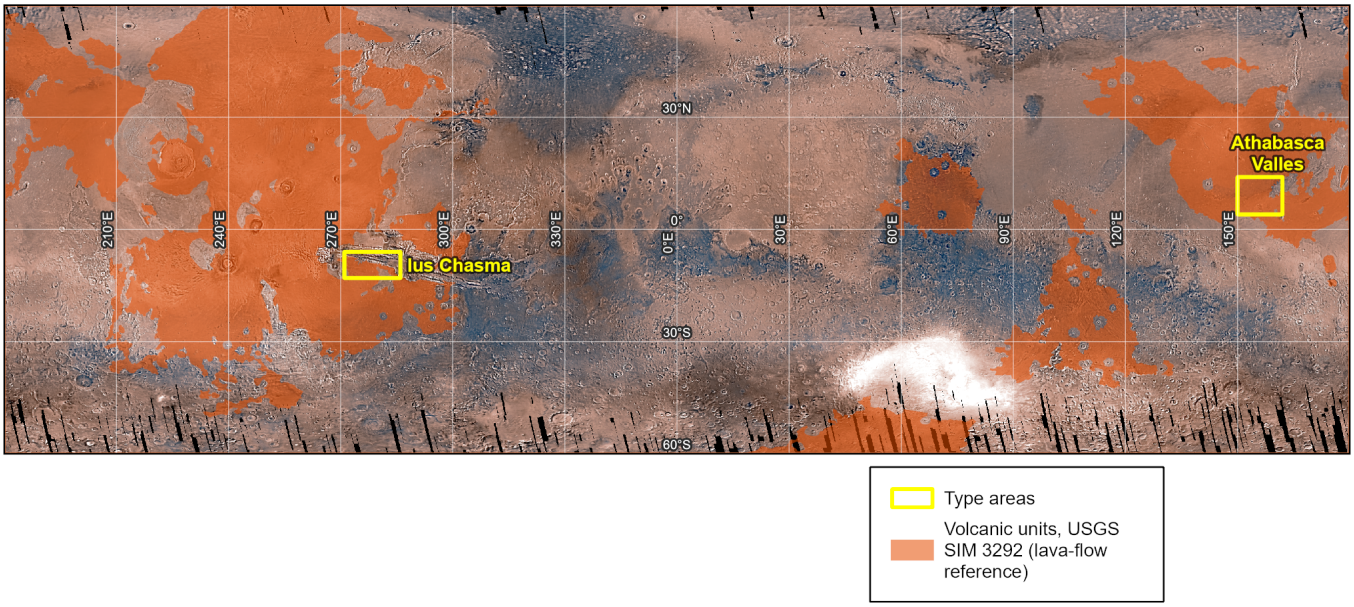

Figure 1. The analysis extent (±60°) over Viking MDIM 2.1 colour, the two type areas, and the volcanic units of the USGS geologic map SIM 3292 (K. L. Tanaka et al. 2014) in orange, the lava-flow reference of this study. Geographic coordinates, centred on 0°E.

radius 3,396,190 m (B. A. Archinal et al. 2018), at 100 m (G100, 213,388 × 71,130, 15.18 billion pixels) and as its exact 2 × 2 aggregation at 200 m (G200, 106,694 × 35,565). The day mosaic sits on the night grid at an offset of exactly +1 and +17,783 pixels, so neither THEMIS mosaic is ever resampled. The DEM cell is 199.99999 m on this sphere, so a nearest-neighbour warp to G200 reproduces 262,144 of 262,144 sampled source pixels exactly, where a cubic warp alters 36% of them by up to 28 m; terrain is therefore computed on G200 and imagery on G100. Slope, aspect and hillshade are computed by Horn's method (B. K. P. Horn 1981) in metres on the projected grid, which removes the vertical-factor warning of the degree-grid products.

The DEM's last column proved defective (nodata in 24,335 rows and wrong values elsewhere; the file also ends 0.46 pixel short of +180°), which painted a 2 km stripe of spurious classes along 180°. It was replaced by the mean of its two neighbours and the affected products were recomputed over that column only.

3.2. *The classification stack*

The seven-band stack for classification is on G200 with 97.33% of pixels valid in every band: Viking R, G and B (bilinear from 231.5 m), THEMIS night and day infrared (2 × 2 averages on their own grid), slope in degrees, and local relief in a 9 × 9 window (1.8 km). Every band is stretched from its 1st and 99th percentiles over ±60° to 1–255. Elevation itself is excluded: a raw elevation band beside 8-bit bands dominates any distance-based classifier, and an earlier map built with one was 62% reproducible from elevation alone. The Viking bands still carry its shading (Section 2).

3.3. *The diurnal-contrast index*

Since calibrated inertia is unavailable, a relative index was formed: each THEMIS band is scaled by its own 2nd and 98th percentiles to $[0, 1]$ and night is subtracted from day, giving a dimensionless value in $[-1, 1]$ that is high for a large diurnal swing (dust-like) and low for a

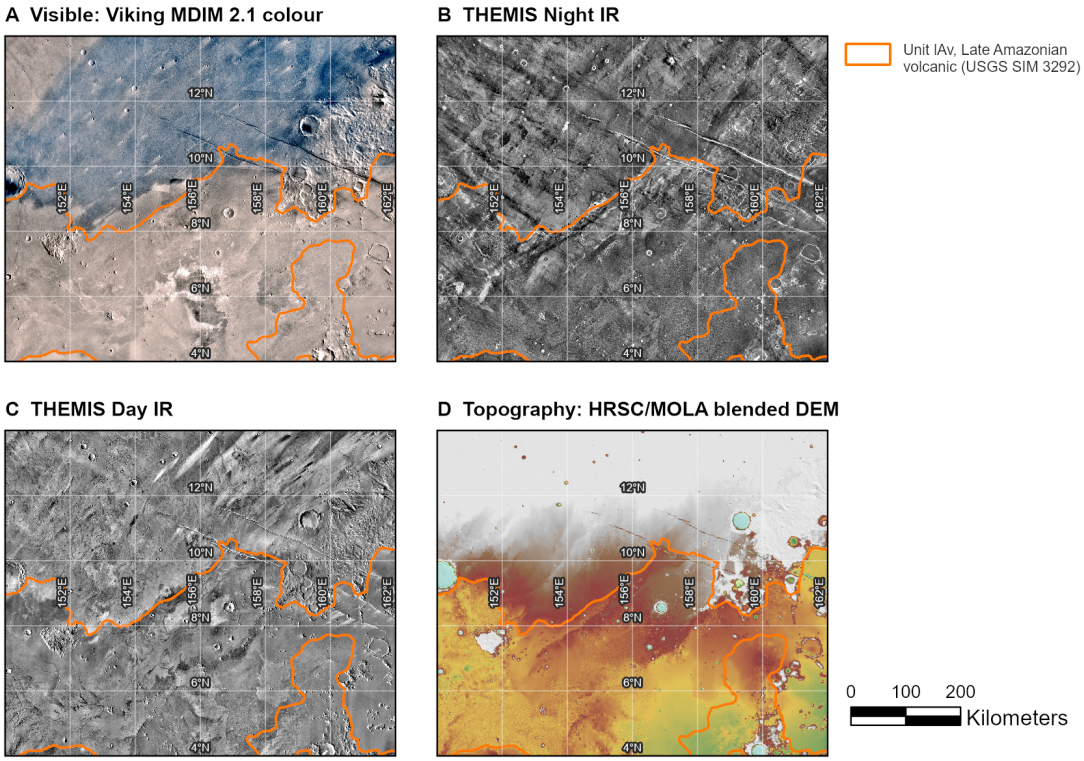

Figure 2. The Athabasca Valles type area on the 100 m stack: (A) Viking colour, (B) THEMIS night infrared, (C) THEMIS day infrared, (D) the HRSC/MOLA elevation model (−4,495 to 1,945 m) over hillshade. The orange outline is unit *lAv*, Late Amazonian volcanic, of SIM 3292.

damped one (rock and coarse debris). It is valid only within the area over which it was stretched. Over Ius Chasma its mean is +0.0757 (s.d. 0.279), and it is nearly independent of topography ($r = +0.108$ with elevation, −0.248 with slope).

3.4. *Closed depressions and channels*

The `Fill` operation (S. K. Jenson & J. O. Domingue 1988) floods every closed depression, which is what a fresh crater is topographically, so fill depth is a crater detector. A depression is kept if it is deeper than 20 m, at least 1 km across (about ten 100 m pixels, the resolving rule used throughout), no more than twice as long as wide, and fills at least 0.55 of its bounding box. The thresholds were tuned on two named craters near Ius Chasma: Perrotin (82.8 km) is found at 77.0 km (−7.1%) and ranks first of 1,321 candidates; Oudemans (124.2 km) is missed, because it is breached into the canyon and its lowest rim point lies 33 m below its floor (Figure 4). Across ±60° the detector runs at two scales: the whole extent at 1.6 km for basins ≥ 20 km, and 8,192-pixel tiles of G200 with a 1,024-pixel (205 km) halo for depressions ≥ 1 km.

Channel centrelines come from D8 flow routing on the filled surface (J. F. O'Callaghan & D. M. Mark 1984; S. K. Jenson & J. O. Domingue 1988) with a 5,000-cell (50 km^2) accumulation threshold, ordered after A. N. Strahler (1957). Because `Fill` turns closed basins into synthetic lakes, only stream cells where the fill raised the surface by at most 1 m are kept.

3.5. *Supervised classification and its score*

Five hundred and twelve polygons were drawn by hand in four landform classes across the planet: 302 Crater, 44 steep or windy hills, 44 lava tube and 122 Normal Ground. Clipped to ±60°, they were split by whole 15° × 15° blocks (14 of 39 blocks held out) into 329 training and 183 held-out polygons (Figure 5), each polygon whole and assigned to the block holding its centre (D. R. Roberts et al. 2017); one held-out polygon overlaps a training one by 51.5 km^2 across a block edge. A support vector machine (C. Cortes & V. Vapnik 1995) was trained on the training polygons (1,000 samples per class) and applied to all of ±60° at 200 m in 21 tiles of 16,384 pixels.

An audit of the 512 labels then revised the scheme to seven classes: Crater narrowed to impact craters, volcanic plains drawn as Normal Ground, and three new classes, volcanic edifice, apron and fluvial channel. Proposals for each label and new polygons for the new classes were built from the geologic map, the Robbins database, the IAU names, the elevation model and the

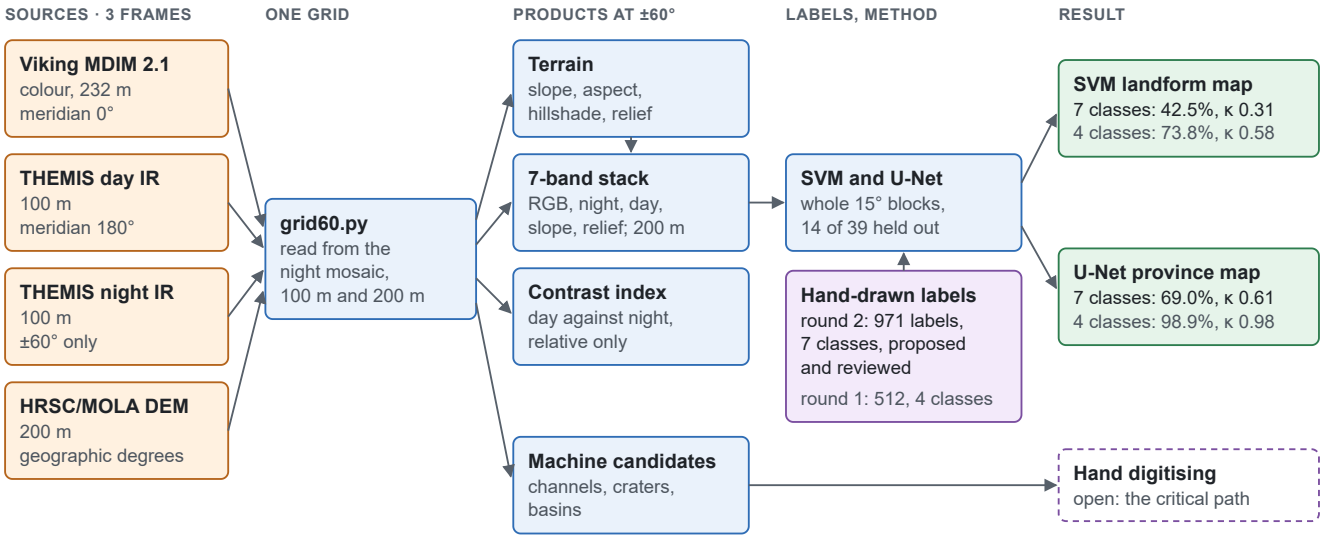

Figure 3. The processing chain. Four mosaics in three frames go onto one ±60° grid; terrain, the seven-band stack and the diurnal-contrast index are built on it, and closed depressions and channels are taken from the elevation model as candidates for review by hand. A support vector machine and a U-Net are trained on hand-drawn labels in two rounds (round 2: 971 labels in seven classes, proposed from the geologic map, the IAU names and the valley map, then reviewed; round 1: 512 labels in four classes; a third label set follows in Section 4.9) and scored on whole 15° blocks, 14 of 39 held out. Scores are area-weighted accuracy and κ on the held-out polygons, 237 for seven classes and 183 for four, so the two lines in a box are not directly comparable.

valley map of G. Alemanno et al. (2018). Every relabel, check and edifice outline was reviewed by eye; the apron and fluvial-strip proposals were screened by a rule written before it ran. On 80 of those proposals, 30 the rule accepted and 10 it rejected in each class, shown on Context Camera images without the rule's verdict, the AI assistant accepted all 27 rule-accepted fluvial strips it decided and 20 of 26 apron pieces (77%, 58–89%), and accepted 4 of 6 and 6 of 8 of the rule's rejects: the screen erred toward strictness. Five of the apron rejects it accepted are the flat fan-shaped deposits of Arsia and Pavonis Montes, which a slope test cannot keep; the third label set admits them by unit, not slope, a net 106 more apron labels. The result, 971 labels, keeps the same 14 held-out blocks (734 training and 237 held-out polygons; the training labels now lie in 129 blocks, not 25, Figure 5), and a support vector machine and a U-Net were retrained on it exactly as before. A rule was written before the run for the seven-class map to replace the four-class one: on the same v2 held-out polygons, (a) its accuracy and κ on the ground both maps can name (v2 classes 1–4) at least the four-class map's, (b) each new class with a user's accuracy above its share of the held-out area, and (c) Crater, hills and Normal Ground each losing under 5 points of producer's accuracy.

A third label set tests whether more and smaller training polygons help. Every seven-class polygon over 1,500 km^2 or wider than 1° was cut on a 0.5° grid, each piece kept only if it passed a written screen for its class, and polygons were added from the crater catalogue, the geologic map and the elevation model, by rule and not by eye, giving 10,269 training and 1,652 held-out polygons in the same 14 blocks. The models were retrained on them; the U-Net's validation loss was unsteady, and the epoch with the lowest, the 19th of 30, was kept. The loss rose late in each later retraining that kept the added polygons, far in all but the one on them alone (to 3.51), and each kept an epoch from before the rise; trained on the cut pieces alone, like every earlier U-Net, it did not rise. Each new map is scored on the 237 held-out polygons, the like-for-like comparison, and on the 1,652. A rule written before the runs recommends a new map over its predecessor only if on both sets its balanced accuracy and κ are at least as high, no class loses 5 points or more of producer's accuracy on the 237 polygons and, for the U-Net, volcanic edifice covers no more of ±60°.

Accuracy is the area-weighted (by cos latitude) share of held-out polygon area given its own class, with Cohen's κ (J. Cohen 1960) and per-class producer's and user's accuracy (R. G. Congalton 1991). Because one class dominates the held-out area, each map is also given its balanced accuracy (the mean producer's accuracy; K. H. Brodersen et al. 2010), the mean of the per-class F1 scores, and its disagreement split into quantity (the wrong amount of a class) and allocation (the right amount in the wrong place) after R. G. Pontius & M.

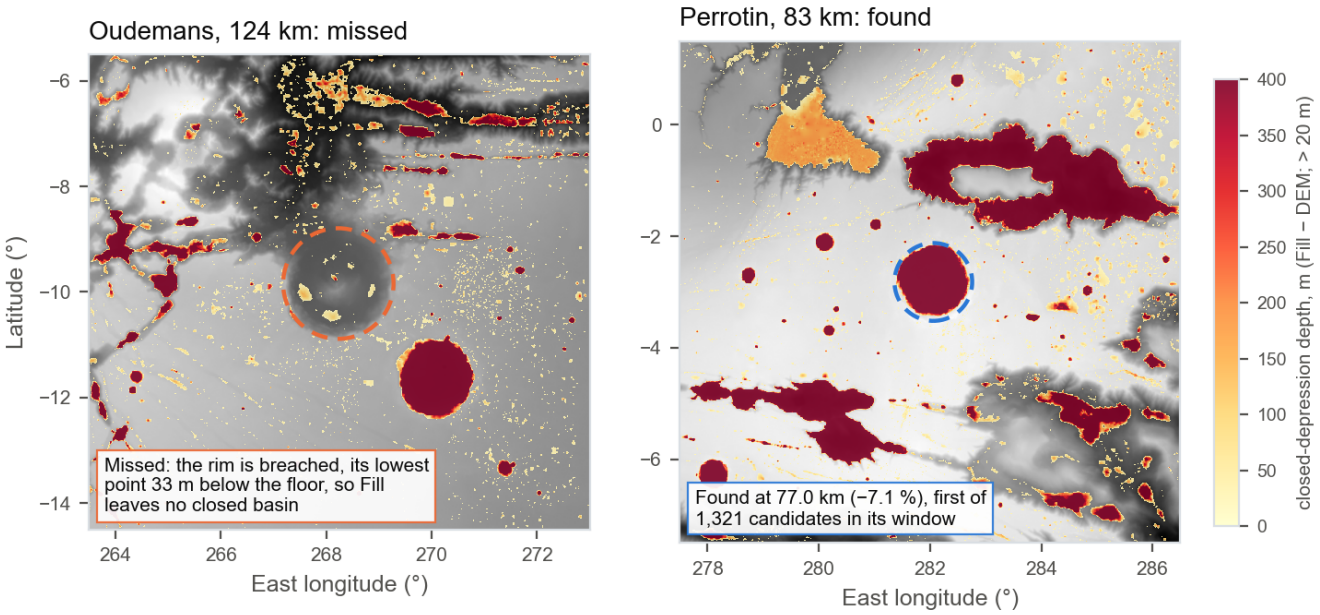

Figure 4. The closed-depression detector on the two named craters it was tuned on. Grey, the HRSC/MOLA elevation; in colour, every closed depression deeper than 20 m by its depth below the filled surface; dashed, the IAU rim. Perrotin is a closed basin and is found at 77.0 km; Oudemans is breached into the canyon, its lowest rim point 33 m below its floor, so filling leaves it empty and the detector cannot see it.

Millones (2011). Five baselines accompany the scores: one class everywhere (58.6% on the held-out set); the class of the nearest training polygon, geography alone; a Gaussian maximum-likelihood classifier on elevation and latitude alone, trained on the same polygons; the share of the map that elevation alone reproduces (each 500 m bin given its majority class); and the share of named craters ≥ 5 km labelled Crater inside $0.7R$ against a ring at 1.5–2.5R. A majority filter removes speckle; its window was set from the resolving rule (about 2 km) rather than from the score, because every score rises with every window when the labelled polygons are large.

3.6. *Deep learning*

Two models were trained on the training blocks only and scored on the same held-out polygons. A RetinaNet detector (T.-Y. Lin et al. 2017) with a ResNet-50 backbone (K. He et al. 2016) was trained on 3,622 chips of $256 \times 256 \times 7$ pixels (stride 128) holding 5,293 boxes; its score is per polygon, recall being the share of held-out polygons at least half covered by boxes of their own class. A U-Net (O. Ronneberger et al. 2015) with a ResNet-34 backbone, class balancing and unlabelled ground ignored in the loss, was trained for 30 epochs on the same chips as pixel labels, in batches of 8 with 10% of the chips held back for validation and the backbone unfrozen, applied to all of ±60° in 243 tiles of 4,096 pixels with a 256-pixel margin, and scored with the same code as the support vector machine. Because the classification tool works internally in 256-pixel tiles, each predicted with 64 pixels of padding, the U-Net was run a second time on a grid offset by 128 pixels, and each pixel was taken from the pass that places it farther from a tile edge. Both models used the Deep Learning Libraries of ArcGIS Pro, built on PyTorch (A. Paszke et al. 2019).

3.7. *Uncertainty*

Intervals are 95% paired bootstrap intervals (B. Efron & R. J. Tibshirani 1993) over whole held-out polygons (1,000 resamples), over contacts for the flow-margin test and over 50 km cells for the channel test, so that spatially correlated pixels are never treated as independent. Polygons in one held-out block share its geology and its errors, so the map scores are also given with block intervals (H. R. Künsch 1989), which resample the 14 held-out blocks (10,000 resamples, every map on the same draw).

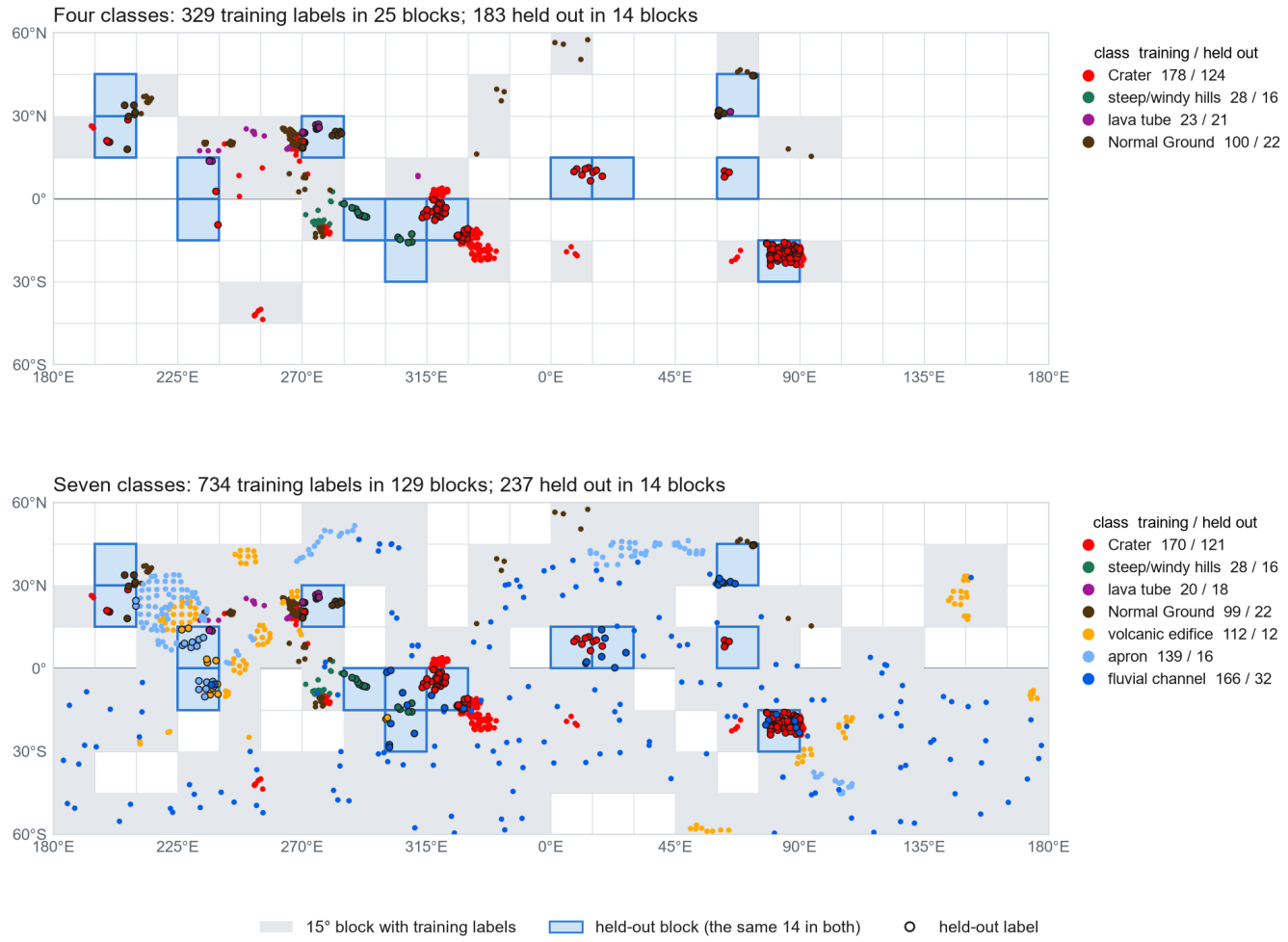

Figure 5. The held-out split. Top: the 512 four-class labels, 329 for training in 25 blocks of 15° (grey) and 183 held out in 14 (blue). Bottom: the 971 seven-class labels, 734 for training in 129 blocks and 237 held out in the same 14. Each dot is one label in its class's map colour; held-out labels are ringed. Geographic coordinates, centred on 0°E.

4. RESULTS

4.1. *The mosaic and the case for fusion*

Figure 6 shows the four layers on the common grid. Over Ius Chasma all five bands are valid on 100% of pixels, and their correlations (Table 2) give the measured case for the fusion: the three Viking bands are one dimension, day infrared is anticorrelated with visible ($r \approx -0.49$; bright dust is cool by day), and night infrared is the most independent band. The effective dimensionality is about three. At Athabasca Valles the pattern repeats (night $|r| \leq 0.15$ against every band). At ±60° the registration was confirmed between Viking red and day infrared: the strongest correlation within ±19 km lies at zero shift ($r = -0.147$), and none survives a 180° roll.

Table 2. Band correlation over Ius Chasma (8,891 × 4,150 pixels at 100 m)

	R	G	B	Day	Night
Viking R	1	0.960	0.925	−0.486	−0.121
Viking G		1	0.993	−0.496	−0.085
Viking B			1	−0.488	−0.068
Day IR				1	0.105
Night IR					1

4.2. *What the thermal pair can and cannot do*

Inside one area the diurnal-contrast index behaves as a material signal should. On the object-based classes of Ius Chasma, built on mean-shift segments (D. Comaniciu & P. Meer 2002), the steep canyon wall (mean slope 21.5°) and the mantled chasma floor (17.0°) are the two

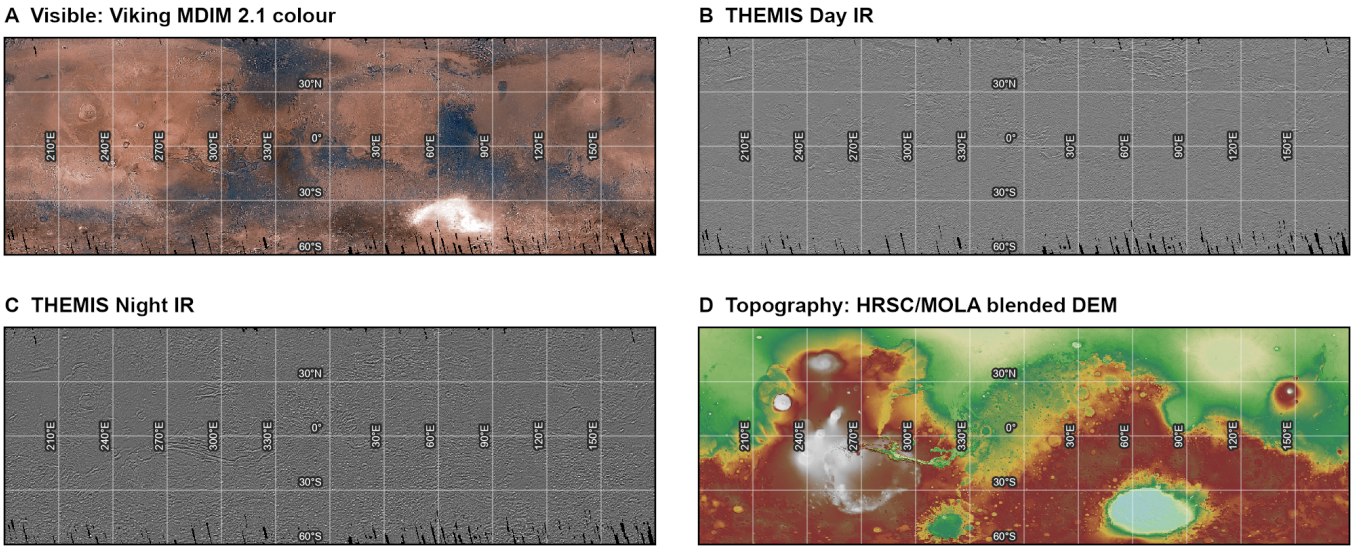

Figure 6. The co-registered mosaic at ±60°, drawn from the 200 m stack: (A) Viking visible colour, (B) THEMIS day infrared, (C) THEMIS night infrared, (D) HRSC/MOLA topography over hillshade. Both THEMIS panels look nearly uniform at this scale because the mosaics are contrast-stretched region by region (Section 4.2).

steepest units, yet differ by 0.46 in the index at opposite signs, and fresh crater ejecta come out damped against the plateau.

Across ±60° the signal does not survive (Figure 7). Over seven provinces whose thermal-inertia contrast is well known (M. T. Mellon et al. 2000; N. E. Putzig & M. T. Mellon 2007), the dusty group (Arabia Terra, Amazonis, Tharsis) and the rocky group (Syrtis Major, Acidalia, Hellas floor, Valles Marineris) differ in the index by −0.0009, against a scatter of 0.13 to 0.23 inside each province. The sources show why: every province reproduces the mosaic-wide THEMIS mean and spread (125.75±34.36 DN by day), so dusty and rocky provinces differ by −0.19 DN by day and +0.02 DN by night, while Viking red, which is not stretched that way, separates them by 47.67 DN. TES measures a fivefold contrast between the same groups (dayside medians of 45 and 246 tiu). The THEMIS DN are local contrast, not radiance.

Calibrated inertia agrees (Figure 8). Averaged into TES cells (∼ 3 km), Viking red tracks log thermal inertia across ±60° ($r = -0.31$ dayside, −0.36 nightside), much less within a 15° block (median −0.06, −0.10): a regional signal, the dust provinces. Night infrared and the index carry a weak local signal in the physical direction (median within-block $r = +0.11$ and −0.08 against nightside inertia, quartiles of one sign) that vanishes planet-wide, as region-by-region stretching predicts; day infrared carries none. In the type areas the index tracks nightside inertia weakly at Ius ($r = -0.22$) and not at all at Athabasca (−0.01), where Viking colour does (−0.64). The prediction that the index would track inertia clearly in both windows fails.

A smaller block does not uncover a stronger signal (Figure 9). In blocks of 15, 10, 5, 2, 1 and 0.5° the pooled within-block correlation of night infrared with log nightside inertia rises from +0.089 at 15° to +0.116 at 2° and falls to +0.032 at 0.5°; the index goes from −0.067 to −0.092 and back to −0.030. The prediction written before the run, $|r|$ of 0.15–0.2 at 5°, failed. Viking red shrinks steadily with the block, from −0.125 to −0.006, as a regional signal should. With each layer shifted 180° in longitude the pooled correlation stays within 0.021 of zero at every size. Below 2° a block holds few TES cells (100 at 0.5°) and a narrower range of inertia, so whether the THEMIS mosaics track inertia below about 100 km cannot be tested with TES.

4.3. *Craters*

The basin pass finds 217,622 closed depressions at 1.6 km in 48 s, 5,144 of them ≥ 20 km, including Hellas (2,697 km across, 7,857 m deep), Utopia (2,093 km) and Isidis (1,026 km). Of the 117 named IAU craters ≥ 100 km inside ±58°, 88 (75%) have a closed depression near their centre and 79 (68%) one whose diameter is within 50%; on those the median diameter error is +2.2% (10th and 90th percentiles −12.6% and +20.8%). This answers H4 with fill depth rather than image gradients.

The fine pass at 200 m, 11 h on the workstation, yields 465,354 closed depressions ≥ 1 km with clean tile seams (density across seams 0.990 and 0.979 of mid-tile). Against Robbins (Figure 10), a candidate matches if its centre lies within half the catalogued diameter and its diameter within a factor of two; a chance baseline

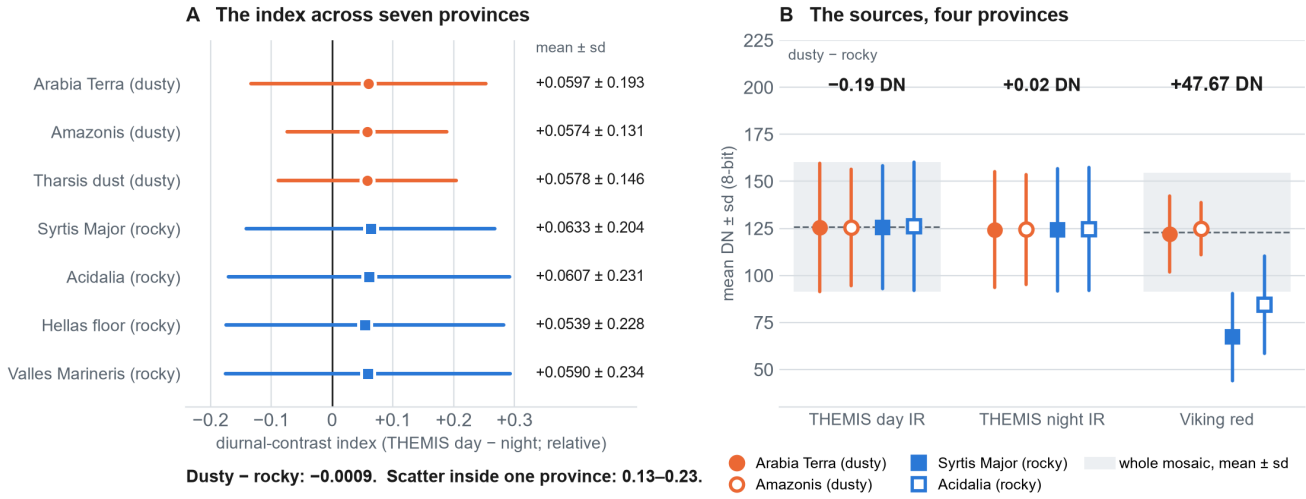

Figure 7. (A) The diurnal-contrast index over seven provinces: dusty and rocky provinces differ by −0.0009. (B) The source DN of four provinces: the THEMIS mosaics repeat the mosaic-wide mean and spread everywhere, while Viking red keeps a 47.67 DN difference.

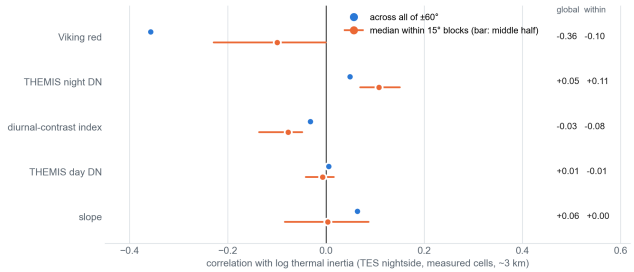

Figure 8. Correlation of each ±60° layer with log TES nightside thermal inertia (N. E. Putzig & M. T. Mellon 2007) in measured cells (∼ 3 km): across the whole extent (blue) and the median within 15° blocks with its interquartile range (orange).

shifts every candidate east by three of its own diameters. Each set is cut at the minimum diameter only where it is scored: recall counts catalogued craters $\geq D$ matched by any depression, precision depressions $\geq D$ matching any catalogued crater. Precision rises from 15.3% at ≥ 1 km to 63.2% at ≥ 5 km and 87.4% at ≥ 20 km, against chance below 14% at every size. Recall levels off near one half (52.4% at ≥ 5 km), and the catalogue's own degradation states (S. J. Robbins & B. M. Hynek 2012) show why: from the most degraded rims to the freshest, 38.4, 49.2, 71.7 and 83.1% of craters ≥ 5 km are found, and fresh rims ≥ 10 km 90–100%; craters whose rim is only partly traced are found at 41.3%, against 63.9% for full rims. Breached and degraded craters are not closed. In the type areas the same rule gives a precision of 11% (Ius) and 13% (Athabasca) at ≥ 1 km.

Mapped by density, the small candidates do not follow the craters: per 0.5° cell their log density correlates with that of Robbins at $r = -0.11$, nearly flat across latitude bands while Robbins doubles, and their pattern shows north–south stripes where the HRSC/MOLA blend is sharpest. Below about 5 km the detector maps the elevation model's resolution as much as the craters. Relative crater density per geologic unit, which needs no detector, ranks the units as expected: on *lAv* Robbins counts 1,122 craters ≥ 1 km per 10^6 km^2, against 1,760 and 2,183 on the older units of the Athabasca window; the detector reverses the order (4,966 on *lAv*) by counting non-crater pits on flat lava. These densities are relative and are not converted to ages (W. K. Hartmann 2005).

4.4. *Channels*

At Ius Chasma, `Fill` raised 21.7% of the scene by more than 1 m and the canyon floor by up to 2,077 m, and 56.1% of the first routed network ran across that synthetic surface (Figure 11). It looked plausible: long, smooth, dendritic trunks down the chasma. With the fill mask, 2,610 candidates (15,428 km) remain; 64% lie on plateau under 2° where routing follows noise, and 188 are steep ($\geq 5°$) and rock-floored (index < -0.15), the subset worth review.

At ±60° the fill mask discards 60.5% of stream cells and leaves 969,822 segments (6.17 million km; orders 1–5: 564,882, 360,236, 41,442, 3,165 and 97; Figure 12). Of 40,410 sampled midpoints, 89.8% lie on ground under 2°, and 96.7% of those of order 3 and higher: the long high-order paths gather on smooth ground where the DEM holds few pits, and their length per cell correlates negatively with the density of 1–2 km depressions ($r = -0.35$). Of the two global valley maps, B. M. Hynek et al. (2010) and G. Alemanno et al. (2018), the

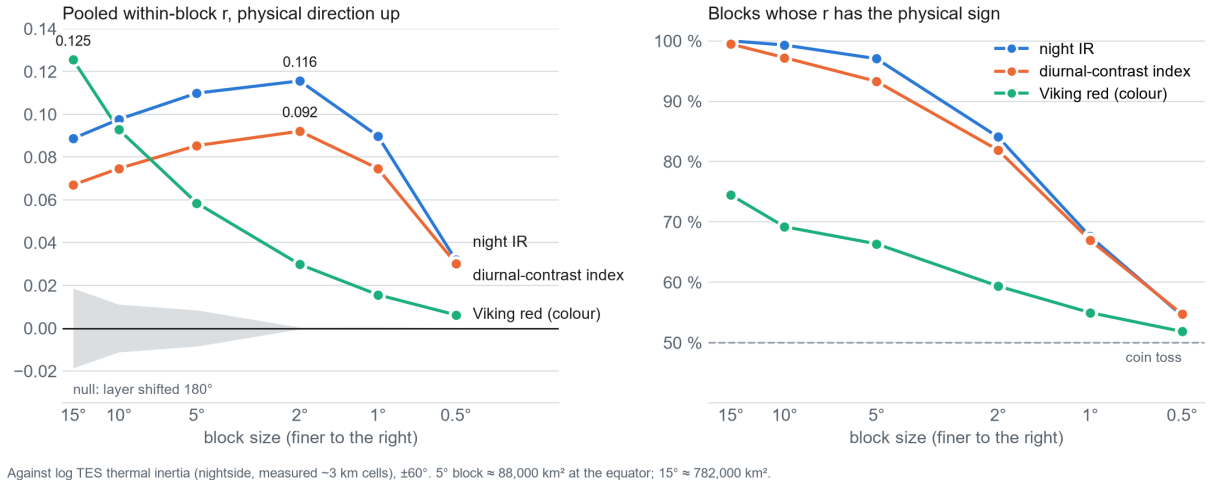

Figure 9. The correlation with log TES nightside thermal inertia against block size. Left: the pooled within-block correlation, the physical direction up (night infrared positive, the index and Viking red negative), and in grey the same with each layer shifted 180° in longitude. Right: the share of blocks whose correlation has the physical sign.

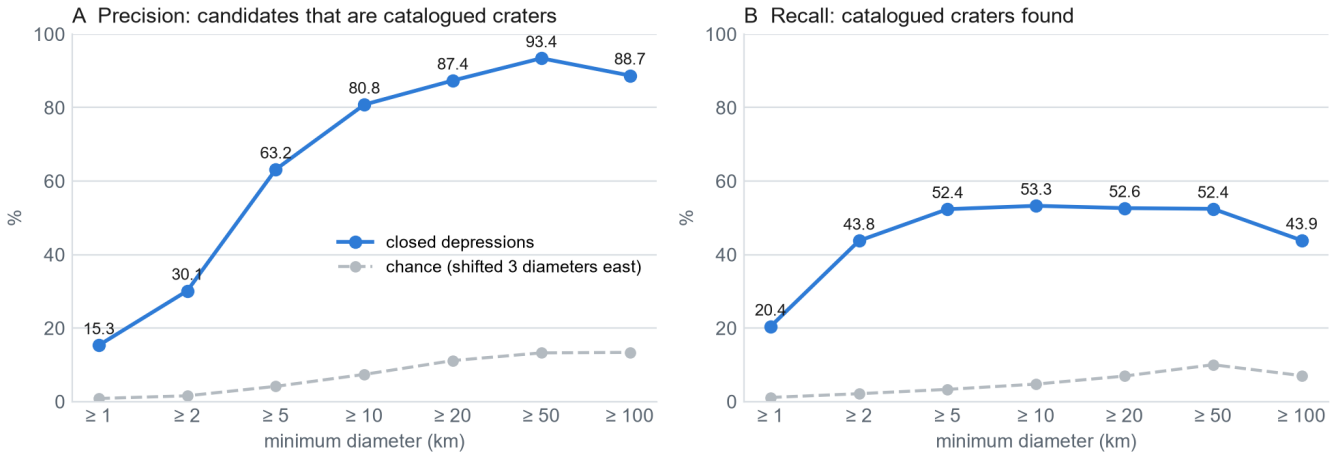

Figure 10. The 465,354 closed depressions ≥ 1 km against the Robbins and Hynek crater database (S. J. Robbins & B. M. Hynek 2012) by minimum diameter: (A) precision, (B) recall, each beside a chance baseline made by shifting every candidate three of its diameters east; each set is cut at the minimum diameter only where it is scored.

second is the reference: against its lines, 2.8% of the order-≥ 3 length lies within 5 km of a mapped valley and 2.0% of the valley length within 5 km of such a segment, no better than the valley map rotated in longitude (precision 1.2 to 4.7%, median 3.3%). Precision is 10.3% for the largest trunks (order ≥ 5) and 18.0% from 45° to 60°S, and 0.1% from the equator to 30°N, where 46% of the order-≥ 3 length lies and few valleys are mapped. The routed network is the elevation model's drainage, not a valley map.

How many tributary orders the 100 m stack resolves (test T7) has no answer the elevation model supports. Over thresholds from 1 to 500 km^2 at Ius Chasma, the highest order falls from 5 to 2, about one order per tenfold rise in the threshold, while only 30.8–36.7% of first-order cells lie on slopes ≥ 2° (53.3% of the window does); the rule set in advance (first-order streams at least half on ≥ 2°) is met by no threshold.

H2 was tested at a first stage with the window standing in for origin (Ius fluvial, Athabasca volcanic). The volcanic window's 3,272 candidates are shallower, not steeper, than Ius's 2,588 (median 1.4 against 10.2 m km^{-1}, difference −8.85 m km^{-1}, interval −15.69 to −5.76), and the thermal z-score separates the two no better than chance (AUC 0.50, against 0.81 for gradient). What separates them is the setting, chasma walls

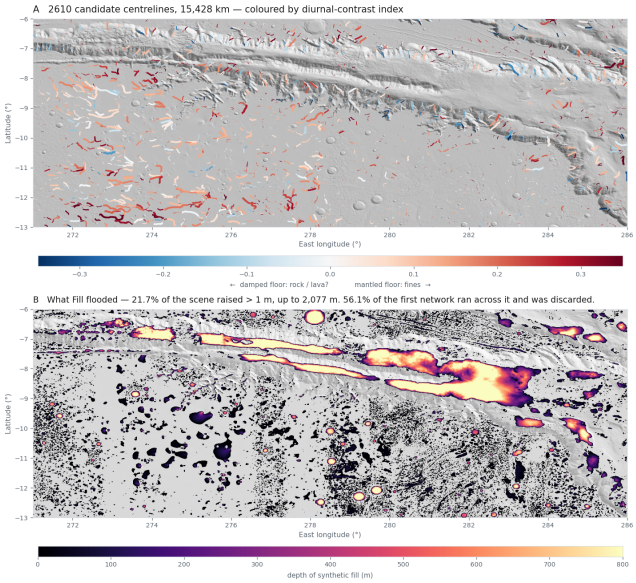

Figure 11. Channel candidates at Ius Chasma. (A) The 2,610 centrelines kept after the fill mask, coloured by the diurnal-contrast index. (B) Where `Fill` flooded the surface: 21.7% of the scene raised by more than 1 m, up to 2,077 m; 56.1% of the first network ran across it and was discarded.

against lava plains, so stage one cannot settle H2. Stage two, the origins compared within one window, needs fluvial and volcanic channels in the same area, and neither type area holds both: Ius Chasma was chosen for water-cut channels, and the outflow forms of Athabasca Valles are draped by lava, their origin debated (D. M. Burr et al. 2002; W. L. Jaeger et al. 2007).

4.5. *Lava flows*

The classification's lava-tube class does not find lava plains. Against SIM 3292 unit groups, expressed as enrichment $P(\text{class} \mid \text{group})/P(\text{class})$ (Figure 13), lava tube is slightly depleted on volcanic plains (×0.84) and enriched on volcanic edifices (×3.30) and aprons (×4.29): steep volcano flanks. Normal Ground covers 76% of the volcanic plains. The other classes sit where the geologic map says they should, so the map is coherent as a terrain map but is not a lava-flow map. A relabelled class trained on the map's volcanic units fared worse: as a fifth class it lowered the other three classes' held-out accuracy from 70.2% to 64.6%, and only 30% of what it called volcanic was, below the 38% base rate. Lava flows in this study therefore come from the geologic map's volcanic units.

H1 was tested on the 14 internal contacts (2,764 km) between the five SIM 3292 units of the Athabasca window. Profiles every 2 km, ±20 km perpendicular to each contact, count as separating when the contrast across them exceeds the 90th percentile of 2,000 random profiles, so chance is 10% (Figure 14). Across the 905 profiles on the ten contacts of *lAv*, day infrared separates 21.0% (interval 3.3–38.4%), Viking red 12.3%, night infrared 11.7% and the index 13.0%. Day infrared leans toward H1 by +8.7 points over Viking red, but the interval (−1.7 to +18.1) includes zero, so by the rule set in advance H1 is not supported. A map at 1:20 M generalises contacts by kilometres, so the test was repeated on lava-flow margins traced on the Context Camera (CTX) mosaic (M. C. Malin et al. 2007; J. L. Dickson et al. 2024): 86 margins (1,476 km) along the *lAv* boundaries, on 126 plates that showed CTX alone, by a rule written before any plate was viewed. The margins were traced by the AI assistant that wrote the project's scripts and read again by it on the same views, blind to the confidence it had given each; no one else has checked them. Across 633 profiles on 78 of them, day infrared separates 19.7% (13.1–27.2%), Viking red 14.8%, night infrared 14.5% and the index 10.0%. Day infrared leads Viking red by +4.9 points, interval −2.7 to +13.1, so H1 is again not supported (Figure 14): the traced margins narrow the interval, 15.8 points wide against 19.8, but leave zero inside it. The second reading kept 56 of the 85 margins it decided (65.9%) as following a flow edge along most of their length, and 81 at least in part; on the 56 day infrared leads by +3.3 points (−7.2 to +13.8), on the 81 by +4.6 (−3.4 to +12.6), so the verdict does not rest on the doubtful lines.

4.6. *The landform classification*

The four-class map (Figure 17) is the 200 m support vector machine map with an 11 × 11 (2.2 km) majority filter. On the 183 held-out polygons it scores a balanced accuracy of 66.7% and 73.8%, κ 0.583 overall, against 58.6% for one class everywhere and 69.9%, κ 0.532 unfiltered (Table 3, Figure 18). Normal Ground is found almost everywhere it was drawn (producer's 99.2%) and Crater is reliable where mapped (user's 93.3%), but lava tube is not a usable class (user's 2.4%). At full resolution the map follows terrain rather than naming landforms (Figure 15). Halving the cell from 400 m bought no held-out accuracy (70.5% raw at 400 m). The map is no longer an elevation map (elevation bins reproduce 55.4%, within a point of its largest class alone, 54.7%), shows no tile seams (ratio 1.00), and labels 56.8% of the ground inside named craters as Crater against 46.2% in the ring around them.

The overall score flatters the map. Crater is 58.6% of the held-out area, and the map's balanced accuracy is 66.7% with a mean F1 of 55.7%, lava tube's F1 being 4.5%. Of the 26.2 points of disagreement, 20.5 are quan-

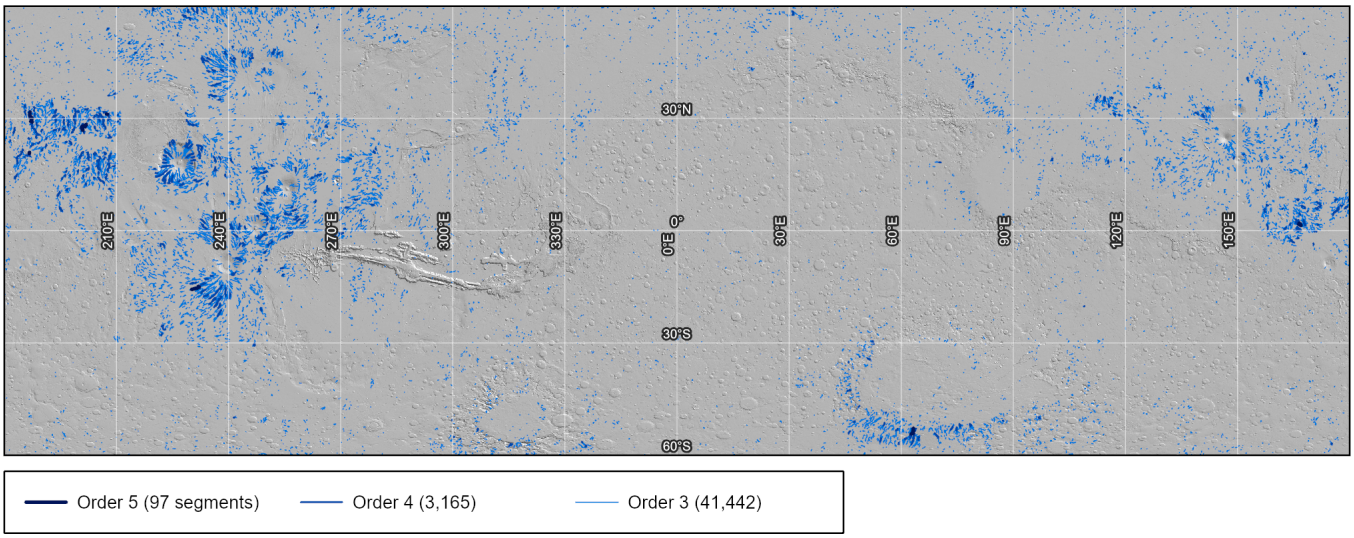

Figure 12. Routed channels of Strahler order 3 to 5 over the ±60° hillshade (orders 1 and 2 are not drawn). Most high-order paths lie on plains under 2° and follow the DEM's makeup as much as real drainage.

	area	Crater	steep/windy hills	lava tube	Normal Ground
Volcanic (v)	22.8%	×0.46	×0.34	×0.84	×1.38
Volcanic edifice (ve)	1.1%	×1.23	×1.19	×3.30	×0.46
Volcanic field (vf)	0.6%	×0.31	×0.34	×1.55	×1.36
Highland (h)	41.1%	×1.34	×0.95	×1.02	×0.79
Highland massif (hm)	2.8%	×1.65	×2.51	×1.79	×0.41
Highland undivided (hu)	2.1%	×1.18	×6.44	×1.22	×0.66
Impact (i)	5.5%	×1.39	×1.56	×1.17	×0.71
Apron (a)	1.3%	×1.02	×4.12	×4.29	×0.31
Transition (t)	7.9%	×1.05	×1.64	×0.71	×1.00
Transition undivided (tu)	2.4%	×0.88	×1.04	×1.28	×1.03
Lowland (l)	9.2%	×0.25	×0.05	×0.04	×1.66
Polar (p)	< 0.1%	×0.39	×0.00	×0.09	×1.57

Enrichment
P(class | group) ÷ P(class)
under ×0.5
×0.5–0.8
×0.8–1.25
×1.25–2
over ×2

Figure 13. The four-class landform map against the unit groups of SIM 3292: how much more often each class occurs in a group than across ±60° (×1: no association).

tity and 5.7 allocation: most of the error is in how much of each class the map paints, not where, giving 38.1% of the held-out ground to Crater against the labels' 58.6% and painting the rest of the Crater polygons lava tube (10.9 points), Normal Ground (8.1) and hills (4.1). Over all of ±60° it is 54.7% Normal Ground, 33.9% Crater, 9.4% lava tube and 1.9% hills.

Geography alone is a stronger baseline than one class everywhere. Giving each held-out pixel the class of the nearest training polygon, and nothing else, scores 83.2%, κ 0.658 overall but 61.3% balanced (Table 3): above the map overall, below it class by class. Shuffling the training polygons' classes among them (20 seeds) never reaches 83.2% (median 47.2%), so where the labels lie carries class information at the scale of the blocks. Resampling the blocks resolves neither gap between the map and this baseline (overall −35.8 to +19.1 points, balanced −16.1 to +16.9); the U-Net clears it overall (+0.9 to +32.6). On the seven-class labels the baseline scores 45.6% and 51.0% balanced, above the seven-class support vector machine on both.

Height is a much weaker baseline. A classifier on elevation and latitude alone, the crustal dichotomy (D. E. Smith et al. 2001) and nothing else, scores 62.5% (38.7% balanced) on the four-class held-out polygons and 31.8% (27.0%) on the seven-class ones, 3.9 and 2.3 points above one class everywhere: what the support vector machine adds over one class everywhere is not elevation, and what beats it is where the labels lie, not how high (Figure 16).

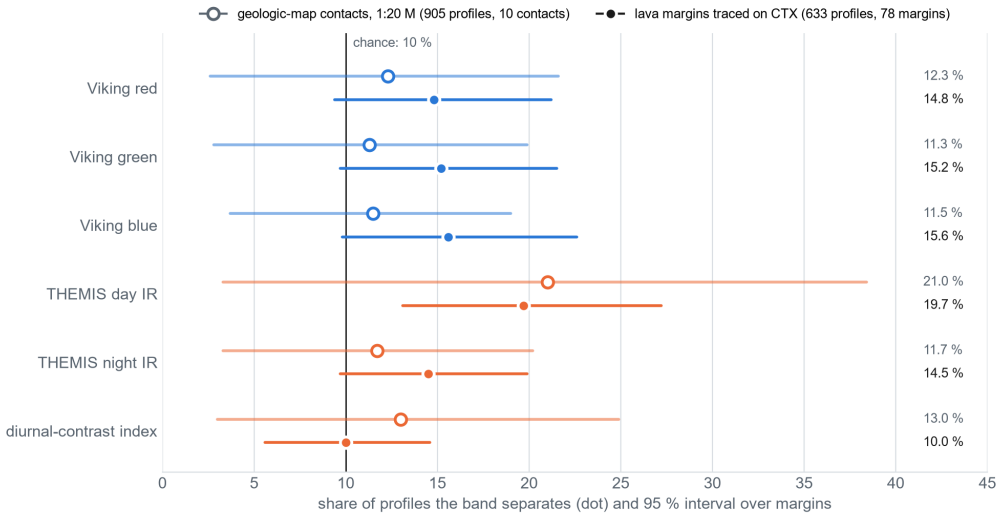

Figure 14. Test T5 (H1): the share of profiles that each band separates across the geologic map's *lAv* contacts at 1:20 M (open) and across 86 lava-flow margins traced on CTX (filled) at Athabasca Valles, with bootstrap intervals over contacts and margins; 10% is chance by construction.

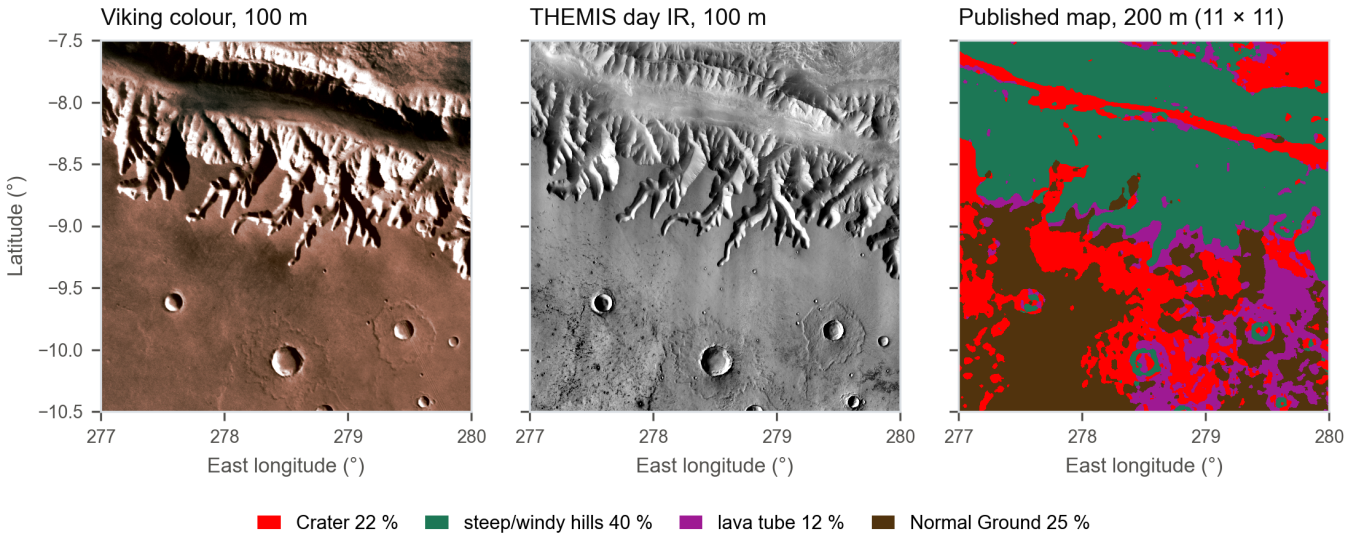

Figure 15. The published map at full resolution, south of Ius Chasma (277°–280°E, 10.5°–7.5°S): Viking colour and THEMIS day IR on the 100 m grid, and the four-class map at 200 m, each map pixel drawn as four image pixels. The map follows terrain rather than naming landforms: the canyon floor is Crater, the walls and spurs hills, the plateau's craters rings of hills, and the plains a patchwork of Crater, lava tube and Normal Ground. Shares are of this window.

Whether the thermal bands help (test T1) was measured twice, with the prediction of little or no gain written first. A Gaussian maximum-likelihood ablation on 2.69 million held-out pixels (Figure 19) finds that the thermal bands add +1.3 points (interval +0.6 to +2.2), that slope and relief alone (68.6%) match all seven bands (67.0%), and that without terrain the visible and thermal bands score below the one-class baseline (42.4%). Retraining ArcGIS Pro's support vector machine over all of ±60° without each band group confirms it (Table 4): the thermal gain is the one difference resolved at both cells, and at 200 m it stays resolved when whole held-out blocks are resampled (+0.46 to +2.67 points). It is not uniform: thermal helps Crater (producer's +1.9 points, nine more polygons won) and hills (+5.5 points) and hurts lava tube (−8.5 points). Weighted by class rather than by area, the gain disappears: balanced accuracy is 63.1% with the thermal bands and 63.4% without at 200 m, 63.5% and 63.4% at 400 m, because Crater and hills hold most of the held-out area. Terrain alone ties

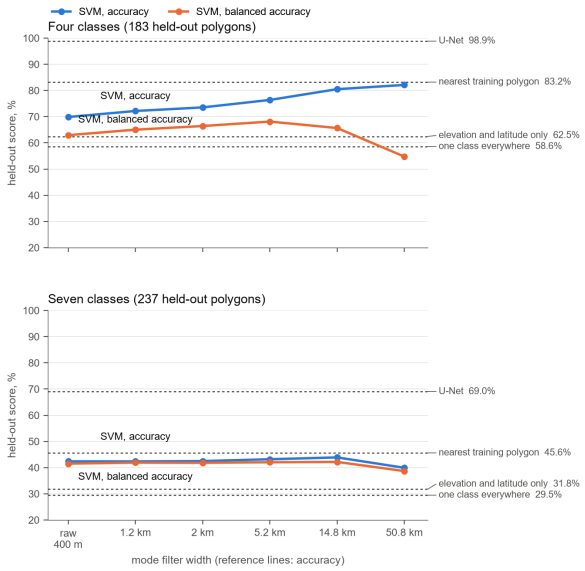

Figure 16. Held-out scores of the support vector machine maps as a majority filter widens from the raw 400 m map to 51 km, four classes above and seven below, against the U-Net, the nearest training polygon's class, a classifier on elevation and latitude alone, and one class everywhere (dashed; accuracy). Filtering brings the four-class map to 82.1%, never above the nearest polygon; the seven-class curve stays flat.

the score but makes a poor crater map (34 of 124 crater polygons won, against 103). The hand-drawn classes are landform classes, a terrain property, and cannot price a material signal.

The seven-class U-Net was retrained once without the two thermal bands, on the third label set, with the prediction written first that it would score within 3 points. It scores 81.2% on the 237 polygons against 87.5% with them (Figure 26), and 78.7% against 86.4% on the 1,652, so the prediction does not hold: the drop is about five times the support vector machine's gain. It is not resolved when the blocks are resampled (−1.4 to +14.8 and −0.7 to +15.8 points): two blocks carry most of it and one goes the other way, and apron, hills and lava tube lose while volcanic edifice gains 10.7 points. The drop is almost all one confusion in one place: apron polygons in the Tharsis block, read as volcanic edifice without the thermal bands, cost 7.75 and 8.34 points. Across the whole of the 14 blocks the two maps differ on 28.1% of the ground, mostly by swaps that leave the class areas where they were. A second training with all seven bands and every setting unchanged, its predictions written first, prices one training against another. It scores 86.8% and 85.5% (−1.6 to +2.4 and −0.9 to +2.4 points from the first), so the score moves by under a point where the bands' removal cost 6.4 to 7.6, and

Table 3. Held-out scores of the landform maps

Map	Acc.	κ	Bal.	F1	Q	A
One class everywhere	58.6	–	–	–	–	–
Nearest training polygon	83.2	0.658	61.3	64.2	14.5	2.2
Elevation, latitude only	62.5	0.236	38.7	–	–	–
SVM 400 m, raw	70.5	0.540	63.5	53.4	23.6	5.9
SVM 200 m, raw	69.9	0.532	63.1	52.8	24.2	5.9
SVM 200 m, 5 × 5	72.0	0.560	65.1	54.4	22.2	5.8
SVM 200 m, 11 × 11[a]	73.8	0.583	66.7	55.7	20.5	5.7
U-Net, two passes	98.9	0.980	94.8	93.9	0.5	0.6
On the 237 v2 held-out polygons						
One class everywhere	29.5	–	–	–	–	–
Nearest training polygon	45.6	0.318	51.0	47.1	28.0	26.4
Elevation, latitude only	31.8	0.160	27.0	–	–	–
SVM four-class[a]	35.7	0.244	42.0	25.1	55.4	8.9
SVM seven-class[b]	42.5	0.310	41.9	38.8	29.1	28.3
U-Net four-class	44.1	0.301	56.7	37.1	55.4	0.5
U-Net seven-class	69.0	0.606	56.9	56.6	14.7	16.3
SVM, third label set[c]	39.1	0.288	45.6	34.8	41.2	19.7
U-Net, third label set[c]	87.6	0.844	84.5	73.7	6.5	5.9

Note: In percent but κ. Acc., overall accuracy; Bal., balanced accuracy; F1, mean per-class F1; Q and A, quantity and allocation disagreement (Q + A = 100 − Acc., to rounding). Area-weighted by cos latitude, scored at 400 m. The first eight rows are on the 183 four-class held-out polygons, the last nine on the 237 seven-class ones; the two sets are not comparable. Nearest training polygon: each pixel given the class of the closest training polygon, geography alone. Elevation, latitude only: a classifier on those two values, F1, Q and A not computed. Block intervals on accuracy (95%, the 14 held-out blocks resampled): SVM 11 × 11 57.5–87.8, U-Net two passes 97.3–99.6, SVM seven-class 23.8–71.5, U-Net seven-class 50.5–80.2, on the third label set 21.3–68.9 and 80.5–95.1. On the probability sample, labelled by the AI assistant (Section 4.10), accuracy by the stratified estimator: SVM 11 × 11 56.9, U-Net two passes 44.4, SVM seven-class 39.9, U-Net seven-class 21.7, on the third label set 40.2 and 49.6. [a]The four-class map of Figure 17. [b]Figure 20. [c]Section 4.9.

apron, which lost 26.3 and 27.3 points of producer's accuracy without the bands, moves by 3.0 and 3.4. Yet its map differs from the first on 26.8% of the ground, nearly the 28.1%, and volcanic edifice moves by 9.8 and 9.9 points, about the 10.7 it gained without the bands. The drop's size and its apron part are larger than one training against another; the edifice gain and the extent of the change are not. A second classifier narrows this further: the scikit-learn stand-in of Section 4.9, given each band's mean and spread over windows to 51 km, was fitted again without the two thermal bands, with the prediction written first that it would move by under 3 points. It falls 8.0 points on both sets (62.0 to 54.0% on the 237 polygons, 59.0 to 51.0% on the 1,652), and the drop is resolved by block (+0.91 to +11.99 and +1.66 to +11.48 points). Apron again loses most (producer's

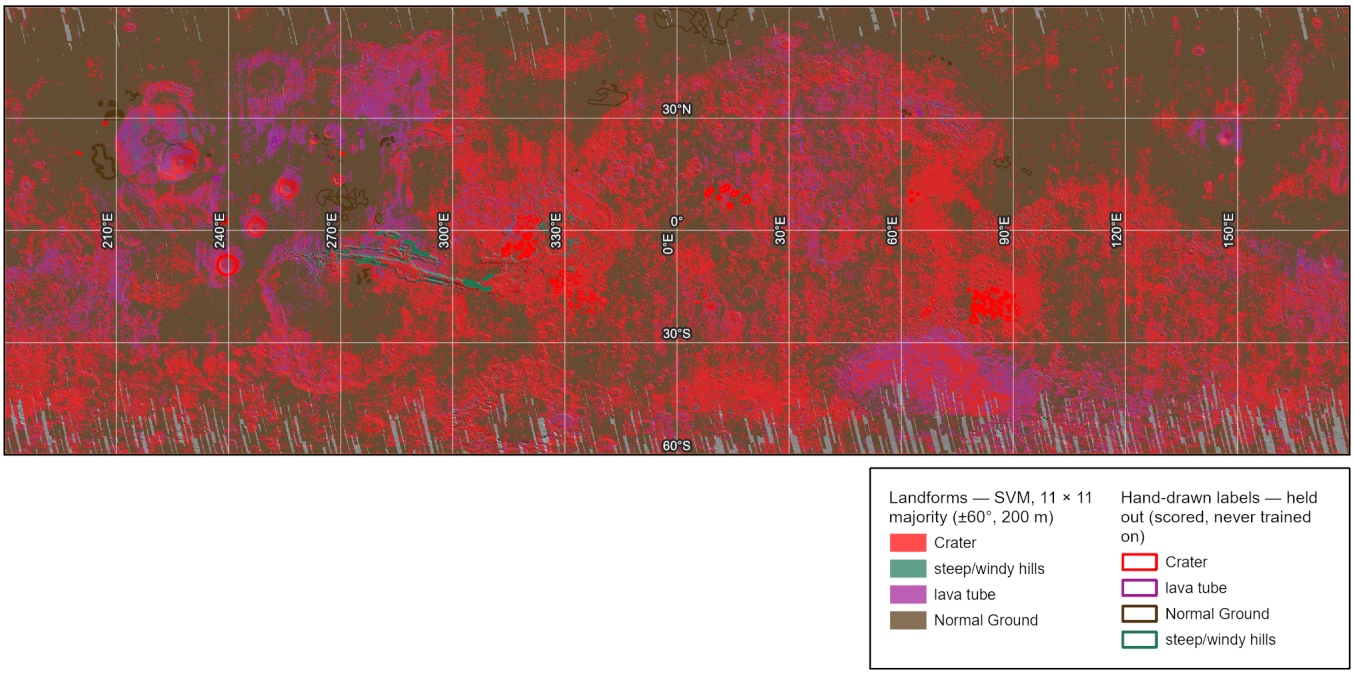

Figure 17. The four-class ±60° landform map: support vector machine on the seven-band 200 m stack, 11 × 11 majority filter, with the held-out hand-drawn labels outlined in their class colours. Held out: balanced accuracy 66.7%; 73.8%, κ 0.58 overall.

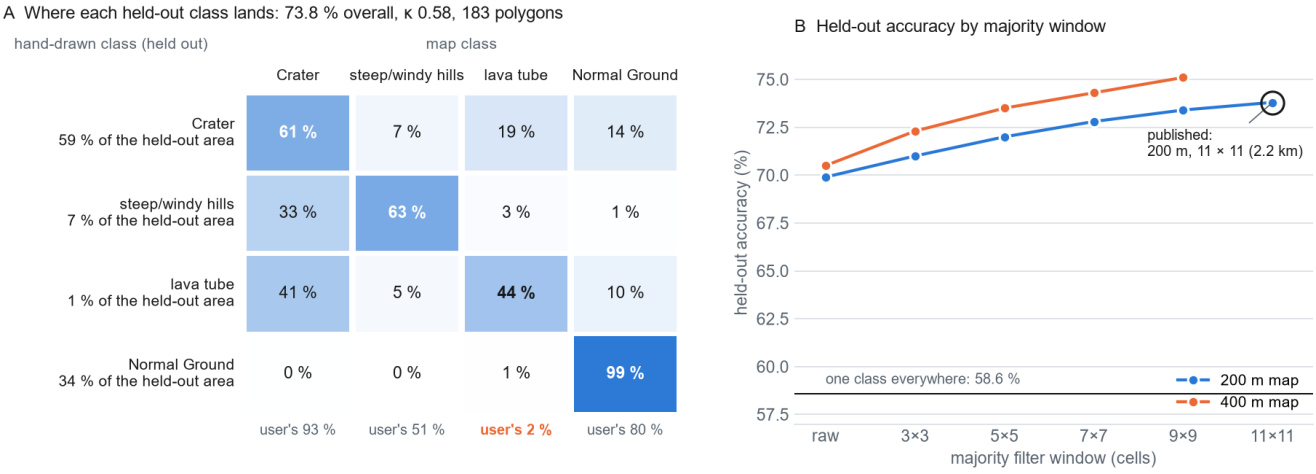

Figure 18. (A) Where each held-out class lands on the four-class map, as a share of its area. (B) Held-out accuracy against the majority-filter window for the 200 m and 400 m maps; the score rises with every window, so the window was set by feature size.

36.15 to 19.74%, 4.79 of the 8.02 points), then volcanic edifice (1.93) and Crater (0.87). Per pixel the thermal bands add under 2 points; seen with their surroundings they carry as much for a support vector machine as for the U-Net. Two more fits, predicted first, place it: their pixel values alone add +0.14 points (−0.79 to +0.81), so the share lies in their statistics over the surroundings; and the point's map coordinates, given in their place, make up 8.67 of the 8.02 points (+2.28 to +13.77), apron returning to 39.90%. On a second draw of the points the thermal bands' loss is 9.94 points, the pixel values' share again nil, and the coordinates make up 7.94 of it. In blocks never seen, where a point lies does most or all of the thermal statistics' work, as a contrast stretch set region by region would; the share is not shown to be material.

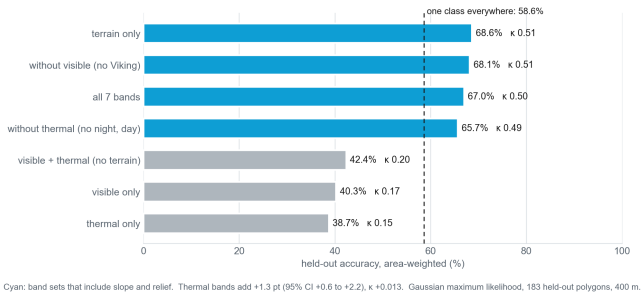

Figure 19. Test T1, first form: held-out accuracy of a Gaussian maximum-likelihood classifier on each band set at 400 m. The dashed line is one class everywhere.

Table 4. Test T1 with ArcGIS Pro's support vector machine

Cell	Left out	Gain (pt)	95% interval
200 m	thermal	+1.36	+0.46, +2.47
400 m	thermal	+1.97	+1.08, +3.35
400 m	visible	−4.26	−10.96, +2.18
400 m	visible and thermal	−0.30	−6.54, +7.73

Note: Gain is the held-out accuracy of the all-band map minus that of the map without the group; paired bootstrap over 183 polygons, 1,000 resamples.

4.7. *Seven classes*

On the 237 v2 held-out polygons the seven-class support vector machine map scores 42.5%, κ 0.310, against 35.7%, κ 0.244 for the four-class map, which cannot name the new classes, and 29.5% for one class everywhere (Figure 20, Table 3). The gain is in overall accuracy and macro F1 (38.8% against 25.1%), not in balanced accuracy (41.9% against 42.0%): the four-class map's error is mostly the wrong amount of each class (quantity disagreement 55.4 points), which naming the new classes removes. By the rule written before the run it would not replace the four-class map: (a) on the ground both can name it scores 66.0% against 79.9% (κ 0.508 against 0.668); (b) apron's user's accuracy, 0.190, is below its share of the held-out area, 0.295, while volcanic edifice (0.261 against 0.194) and fluvial channel (0.346 against 0.065) pass; and (c) Crater loses 25.7 points of producer's accuracy and hills 12.4. The new classes find their units of the geologic map: volcanic edifice covers 50.8% of the volcanic edifice units, 4.15 times its share of ±60°, and apron 36.7% of the apron units, 5.69 times. Fluvial channel, however, takes 33.0% of ±60° against 6.5% of the held-out label area, and lava tube's producer's accuracy falls from 0.665 to 0.063.

Seven classes are nevertheless the scheme this paper presents. They separate impact craters from the volcanic shields that the four-class Crater class took in, and name volcanic edifices, aprons and fluvial channels, which four classes cannot; the cost, measured above and in Section 4.8, falls on the original classes. The four-class maps are kept beside them, and the maps the paper leads with are those on the third label set (Section 4.9).

4.8. *Deep learning*

The RetinaNet detector found 41.9% of held-out Crater polygons with 87.6% of its in-polygon boxes correct (always naming Crater gives 71.1%) and 22.7% of Normal Ground polygons at 56.9% precision, but never named hills or lava tube (Figure 21, left). Retrained with the backbone unfrozen for 30 epochs and with extra chips of the rare classes, it named hills (25.0% recall, 100% precision) but lost Normal Ground (4.5% recall) and over-predicted lava tube on craters; by the rule written before the run it did not improve. Boxes suit compact objects (C. Lee 2019) rather than area classes, and the whole-chip boxes of plains limit this model.

The U-Net is a different matter. In the held-out windows it scored 98.75%, κ 0.977, and over all of ±60° 98.8%, but with seams at its 256-pixel tiles (phase ratio 8.6–9.2 against about 1.0 for the support vector machine maps). The second, offset pass lowered the ratio to 1.2 and changed nothing else measured: 98.9%, κ 0.980 held out, with lava tube at 77.1% user's accuracy against 2.4% (Figure 21, right, and Figure 22). The map is a map of terrain provinces, not of features: the southern highlands are Crater, the northern lowlands Normal Ground, lava tube lies on Tharsis and Elysium and hills along Valles Marineris; it makes 55.1% of ±60° Crater and 39.3% Normal Ground, where the support vector machine map makes 33.9% and 54.7%. It labels 91.3% of the ground inside named craters as Crater, but also 82.6% of the ring around them, and elevation bins reproduce 68.0% of it. The held-out labels mark areas that are mostly one class at 50 km, and a model that sees 51 km of context reproduces those areas; the score measures agreement with the hand-drawn areas at province scale. Of the four-class maps the support vector machine map is the landform map, with the U-Net map beside it as a province map.

Retrained on the seven classes (17,444 chips, 6.2 h), the U-Net scores 69.0%, κ 0.606 on the v2 held-out polygons, against 44.1%, κ 0.301 for the four-class U-Net and 42.5% for the seven-class support vector machine (Figure 23). Over the four-class U-Net, as for the support vector machine, the gain is in overall accuracy and macro F1 (56.6% against 37.1%), not in balanced accu-

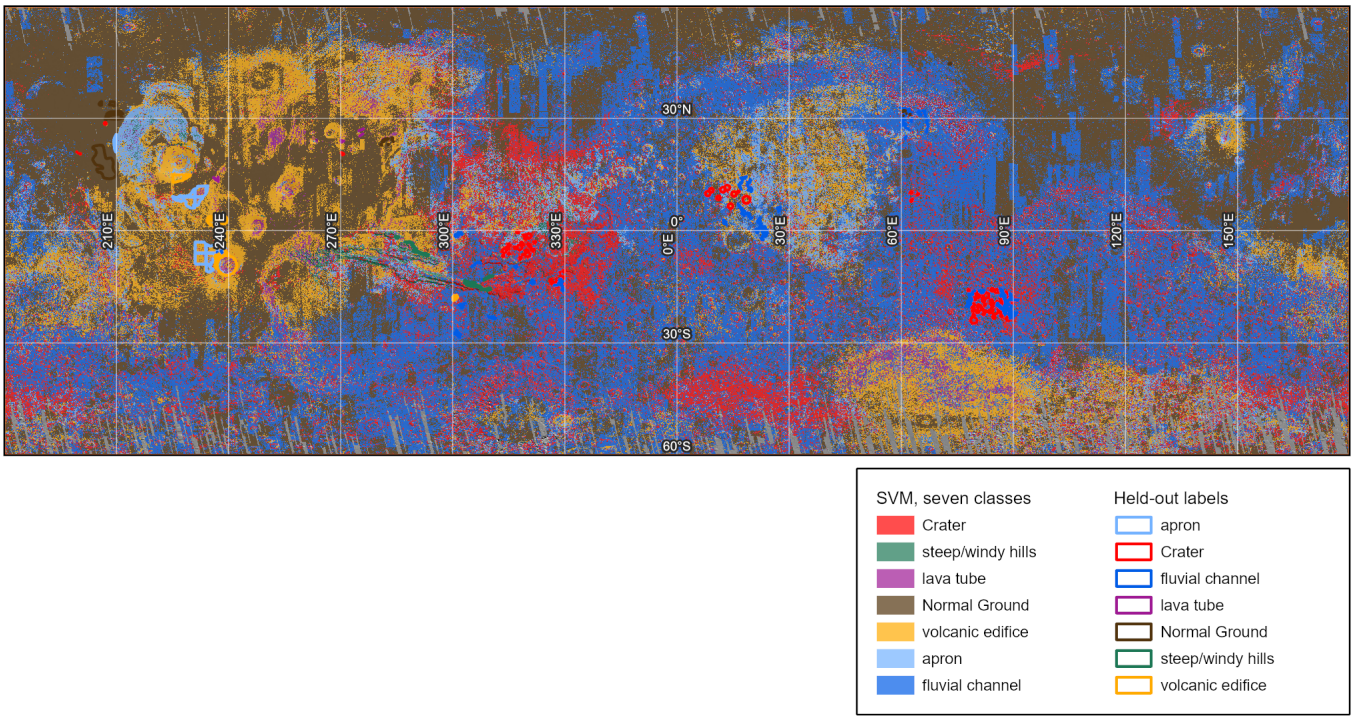

Figure 20. The seven-class ±60° landform map on the reviewed labels: the same support vector machine and stack as Figure 17, trained on the reviewed labels, with the 237 held-out polygons outlined in their class colours. Held out: 42.5%, κ 0.31.

racy (56.9% against 56.7%). Its lead over the seven-class support vector machine is resolved by balanced accuracy on these polygons, 56.9% against 41.9% (block interval +2.3 to +19.5 points), but not by accuracy (−9.7 to +47.8): most of the 26.5 points lie in the two Tharsis blocks, 47.5% of the held-out area, where the support vector machine scores 16.0% and 25.6% and the U-Net 68.6% and 96.3%. On the 1,652 held-out polygons of the third label set, in the same blocks, both maps score about 3 points lower (65.8% and 39.8%) and the order holds, but the balanced-accuracy lead, 52.9% against 40.4%, is no longer resolved by block (−3.7 to +16.3 points): the lead is clear by polygon and rests on a few blocks. The four-class U-Net's lead over the support vector machine map holds by block (+11.1 to +41.0). Each new class has a user's accuracy above its share of the held-out area: volcanic edifice 0.487, apron 0.798 and fluvial channel 0.702. By the same rule it would not replace the four-class U-Net: on classes 1–4 it scores 61.9% against 98.9%, and Crater, hills and Normal Ground lose 41.3, 52.6 and 28.8 points of producer's accuracy. Its fault is volcanic edifice, which covers 40.0% of ±60° against 19.4% of the held-out label area and takes 71% of the lava-tube area. Its Crater class is more crater-shaped than the four-class one (0.524 inside named craters against 0.241 in the ring around them), and elevation alone reproduces 48.4% of it.

The named-crater baseline extends to two of the new classes, on names with no label at their centre; as the edifice and fluvial labels were drawn from the same names and valley map, it shows consistency with the labels' sources, not accuracy. Fluvial channel is more common within 2 km of the valley lines of G. Alemanno et al. (2018) than 10–20 km from them in 83% of 83 named valles on the U-Net map (0.222 against 0.101) and in 66% on the support vector machine map (0.402 against 0.353); shifting either map by 180° in longitude brings this to chance. Volcanic edifice does not concentrate on named edifices: it is more common inside $0.7R$ than in the ring for none of 10 smaller named Mons, Tholus and Patera on the U-Net map and 3 on the support vector machine map. Seven of the ten lie on highland, basin or transition units, and on the three volcanic ones both maps give the ring more edifice than the summit; the large volcanoes carry labels and cannot be checked this way.

4.9. *The third label set: the lead maps*

On the 237 polygons the U-Net on the third label set scores 87.6%, κ 0.844, balanced accuracy 84.5%, against 69.0%, 0.606 and 56.9%; the support vector machine scores 39.1%, κ 0.288, balanced accuracy 45.6%, against 42.5%, 0.310 and 41.9% (Table 3, Figures 24 and 25). By block the U-Net's difference runs from +13.0 to +38.7 points of balanced accuracy, which is resolved; the sup-

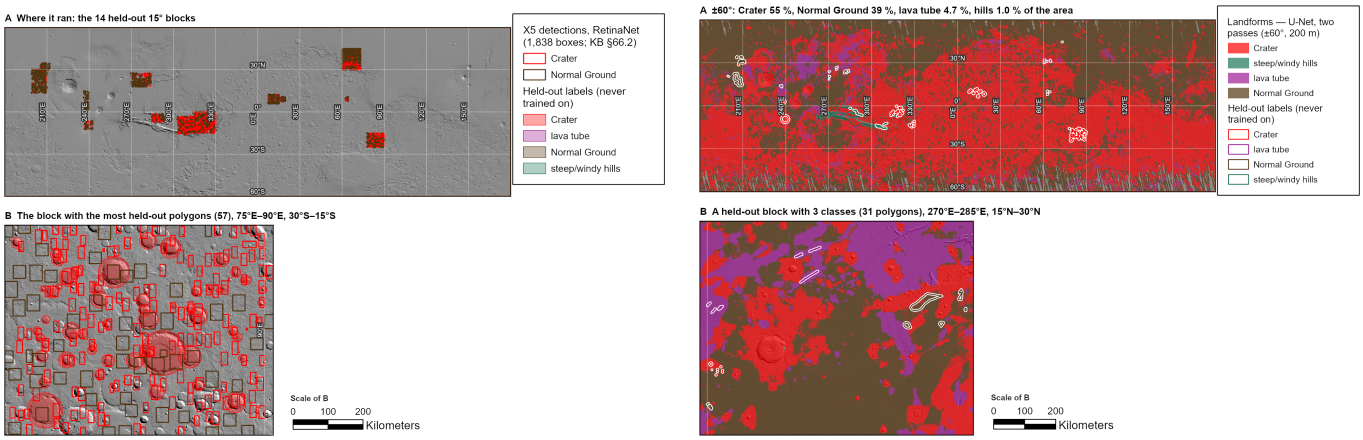

Figure 21. Left: RetinaNet detections in the 14 held-out windows, with the block holding the most held-out polygons (57, at 75–90°E, 15–30°S) enlarged. Right: the two-pass U-Net map of terrain provinces over ±60°, with the held-out block holding the most classes (270–285°E, 15–30°N) enlarged.

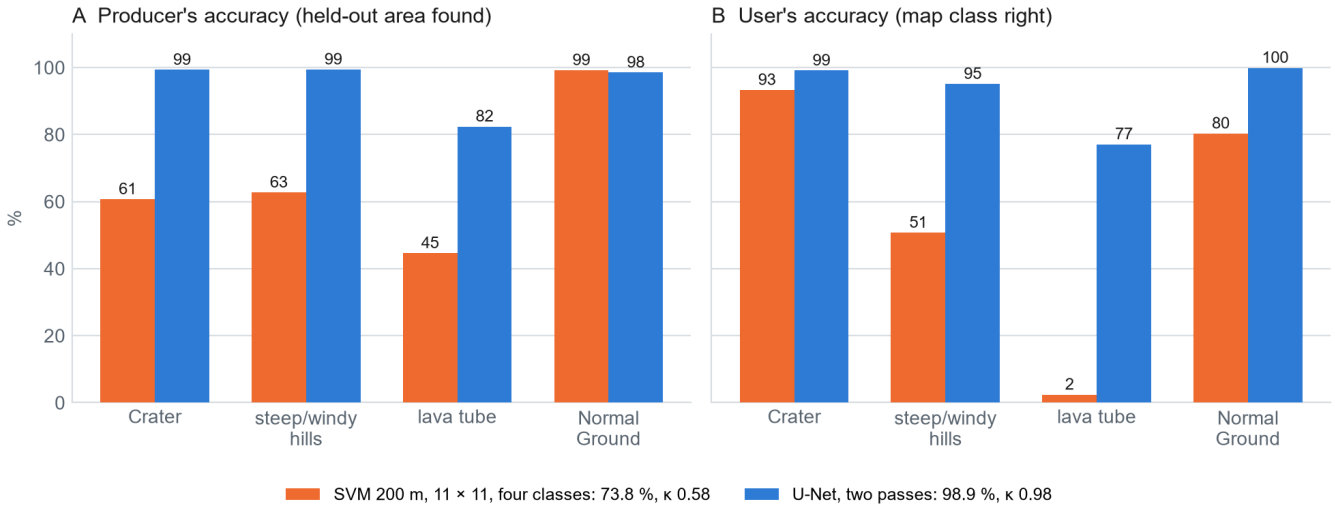

Figure 22. The four-class support vector machine map and the two-pass U-Net map under the same scorer: producer's (A) and user's (B) accuracy per class on the 183 held-out polygons.

port vector machine's, −2.7 to +8.7, is not, nor is its fall in accuracy (−9.7 to +2.3). On the 1,652 polygons they score 86.5% and 36.5%, against 65.8% and 39.8%. The U-Net leads the nearest training polygon by +31.3 to +51.8 points of accuracy on the 237 polygons; the support vector machine does not separate from it (−24.9 to +24.3). Volcanic edifice covers 18.0% of ±60° on the new U-Net map, against 40.0%, and the two new maps agree on 47.5% of ±60°, against 39.7%.

The score was above the range registered for it (60–78%), so a leak was sought first: no held-out polygon shares measurable area with a training polygon, where the same test finds 89.2% of the second label set's training area under the third's. The gain is broad: 13 of the 14 blocks rise and 183 polygons gain against 38 that lose. Of its 18.6 points Crater gives 8.6 and Normal Ground 5.6, as the earlier model's calling Crater edifice or apron, and Normal Ground edifice, nearly stops. The Tharsis block with 29.7% of the pixels gains least and reads 77.2%.

Two retrainings set out before they ran split the U-Net's gain (Figure 26). Trained on the cut pieces alone (7,401 polygons), adding nothing, it reaches a balanced accuracy of 78.1% on the 237 polygons; on the second label set plus the added polygons alone (3,602), cutting nothing, 77.6%; against 84.7% with both and 56.9% with neither. Each half gives part of the gain, about equally (their difference, −7.8 to +5.8 points, is not resolved by block), and the two together most. By pixel accuracy the added polygons alone score highest, 89.3%, but lose hills (−37.4 points) and lava tube (−38.6): most of their hills are the rule's.

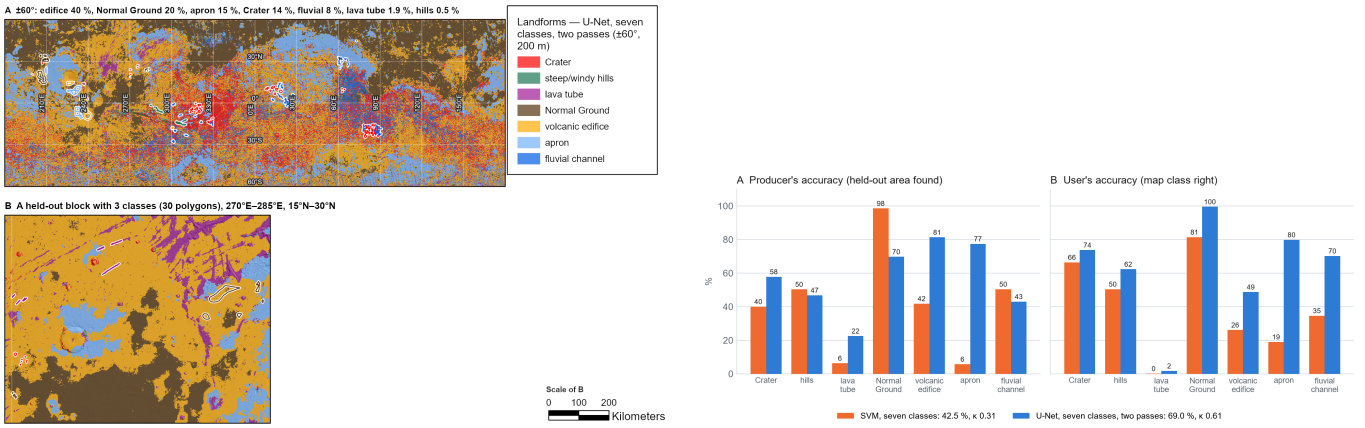

Figure 23. Left: the seven-class U-Net over ±60°, with the held-out block holding the most classes (270–285°E, 15–30°N) enlarged. Right: the seven-class support vector machine and U-Net maps under the same scorer, producer's (A) and user's (B) accuracy per class on the 237 v2 held-out polygons.

The support vector machine's flat score has another cause (Figure 27). A stand-in, a scikit-learn support vector machine on the same seven bands scored at 30,000 points in the 237 polygons, repeats its direction: 39.3% trained on the second label set, 38.8% on the third, at 1,000 pixels a class. Ten times the 1,000 pixels a class that Pro draws changes its accuracy by −0.6 to +1.0 points, and the third label set does not raise it (−7.5 to +3.1): what holds the support vector machine is one pixel's seven bands, not its sample or its labels. The third set moved its boundary between Crater and hills. The rule's hills, half of them on crater flanks, are harder to tell from Crater by those bands (two-class accuracy 0.722 against 0.877), and Pro's map on that set reads 21.0% of the held-out Crater as hills, against 4.9%: Crater, the larger class, loses what hills gains, so accuracy falls while balanced accuracy rises. Given each pixel's surroundings as well, each band's mean and spread over 1, 5 and 20 km, the stand-in rises from 39.2% to 57.1% on the third label set (+9.7 to +28.2 points) and to 60.9% on the second, so the third set still does not help it (−9.6 to −0.2); the U-Net, at 88.2% on the same points, leads it by +12.0 to +42.9. Over 51 km, the U-Net's own view, the stand-in reaches 62.0% (+1.85 to +7.3 points more) and the U-Net still leads by +9.7 to +37.1, while the third label set still does not help it: what the U-Net draws from that set is learnt spatial pattern, not band statistics over its view. Of its 26.2-point lead at those points, apron gives 14.85 and Crater 8.48, classes defined by where they lie, around an edifice or inside a rim; the stand-in does better on hills and fluvial channel. Given the point's map coordinates as well it reaches 69.2%, and the U-Net still leads by 19.0 points.

The same 14 held-out blocks have scored every model in this paper. In a run set out before it ran, 14 other blocks drawn at random, holding 895 polygons of the third label set, were held out instead and both models retrained on the other 9,366 polygons. The U-Net scores 86.6%, κ 0.815, balanced accuracy 72.2% (block interval 74.8–94.6%), as the published model does on the 1,652 polygons, and leads the nearest training polygon by +5.0 to +49.7 points; the support vector machine scores 37.9% and does not separate from that baseline (−51.1 to +14.6). A second draw, 14 blocks holding 1,602 polygons and sharing one block with the first, gives the U-Net 85.4%, κ 0.781, balanced accuracy 69.2%, and the support vector machine 45.6%, again not separated from the baseline; but there the nearest training polygon scores 80.6%, since 43.5% of those polygons are apron, which it gets right 92.7% of the time, and the U-Net's lead over it is resolved by balanced accuracy (+3.8 to +16.7) and not by accuracy (−2.5 to +26.7), though it leads in 13 of the 14 blocks. That is the draw's class mix: weighted to the class mix of the 1,652 polygons, the baseline falls to 71.9% there and the U-Net's lead is resolved (+3.3 to +27.9), and over both draws' 28 blocks it is resolved as scored (+0.7 to +30.9). So the U-Net's score is not a property of the first 14 blocks. On the first draw its hills (18.3% producer's accuracy, 64.5% on the second) and fluvial channel (37.4%, 34.4%) are weaker. Every hills polygon there is one the rule placed, and on the first 14 blocks such polygons score 35.2% against 81.8% for pieces of the author's, so the model has learnt the author's hills more than the rule's. The rule kept its hills clear of craters of 10 km or more only by 1.3 radii, and half of its training hills lie within two radii of one, against 6.5% of the author's; those the model reads as Crater (79.4% and 81.3% of the held-out ones within two radii, 28.2% and 27.3% beyond), and more of the fresh blocks' hills lie that close. Fluvial channel falls in

one block, 6_−3, which holds 66 of its 147 polygons at 10.9%.

By the rule written before the runs the U-Net map replaces its predecessor. The support vector machine map does not: its κ falls on both sets and Crater loses 21.8 points of producer's accuracy (40.0% to 18.2%), while hills gain 38.3 and fluvial channel 12.7, and it keeps fluvial channel over 31.9% of ±60°. Both maps on the third label set are presented all the same, as the pair the paper leads with, the second against that rule.

Each map has its checks. On 469 named craters the U-Net's Crater covers 93.5% of the inside and 53.0% of the ring around them, against 61.3% and 24.8% for its predecessor; the support vector machine's, 21.1% and 6.6%. Fluvial channel lies nearer the valley lines than 10–20 km from them in 94.1% of 85 named valles on the U-Net map and 68.2% on the support vector machine map. Volcanic edifice is still denser inside none of 10 smaller named edifices on the U-Net map, and 3 on the other, and only 30.5% of the U-Net's edifice lies on the volcanic units; lava tube's user's accuracy is 8.0% on the 237 polygons and 18.3% on the 1,652. A larger majority filter does not help the support vector machine: its accuracy falls from 37.4% raw to 36.3% at 13 pixels and 33.2% at 127.

4.10. *A probability sample*

A probability sample (P. Olofsson et al. 2014) gives scores free of where the polygons were drawn. Twenty-five points were drawn in each class of the seven-class support vector machine map, with probability proportional to area, and 35 more inside the author's held-out polygons were hidden among them as a check on the labeller; the AI assistant labelled all 210 blind on Context Camera images and elevation-model shading, with no map class shown. On the hidden points it agreed with the author's classes on 66.7% of the 27 it decided ($\kappa = 0.604$), short of the 70% set in advance, and missed the apron and fluvial polygons, so what follows is the assistant's reading, reported at the author's decision. On the 142 sample points it decided, by the stratified estimator, the published four-class map scores 56.9% (48.8–65.0%), the four-class U-Net 44.4% (35.8–52.9%), the seven-class support vector machine 39.9% (35.6–44.2%) and the seven-class U-Net 21.7% (13.7–29.7%), each below its score on the held-out polygons. On the same points the seven-class U-Net minus the seven-class support vector machine is −18.3 points (−26.7 to −9.9), the reverse of its lead on the polygons, but the labeller's lean to Normal Ground favours the support vector machine, which maps more of it. On the 57 points not read as Normal Ground the difference is −4.1 points (−27.9 to +19.8), so the sample does not rank those two maps. Normal Ground covers 79.8% of the sampled ground (73.4–86.3%) against 33.3% of the seven-class map, and fluvial channel, a third of the map, 3.3% (a user's accuracy of 4.8%). The labeller named none of the 175 sample points apron, its weakest class on the check, so the apron share is not measured.

The points were drawn before the third label set, so they are not its own. The U-Net on it scores 49.6% (40.8–58.5%), +28.0 points on its predecessor (+17.5 to +38.4), and the support vector machine 40.2% (34.3–46.1%), +0.3 (−5.4 to +5.9). The lean does not explain the U-Net's gain: on the 57 points not read as Normal Ground it scores 56.4% (36.8–75.9%) against 28.7%, a difference of +27.7 (+12.3 to +43.1). The U-Net minus the support vector machine is +9.5 points on all the points, which is not resolved (−0.7 to +19.6), and +34.4 on those 57, which is (+5.1 to +63.7). The U-Net stays far below its polygon scores; the support vector machine's sample score is level with its own.

5. DISCUSSION

5.1. *The hypotheses*

Table 5 collects the verdicts. None of the four hypotheses is supported as stated. H3 failed outright: Iso Cluster partitions trained on the west and east halves of Ius Chasma agree less for the four-band composite (adjusted Rand index 0.40, L. Hubert & P. Arabie 1985) than for any single input (0.61 to 0.89), although the composite gives the most coherent maps; a one-band clustering is close to a density slice and reproduces easily. H1 was tested on the geologic map's contacts and on margins traced on CTX, and is supported on neither; H2 was tested on stand-ins (windows) and cannot be taken further in these type areas, neither of which holds both channel origins. A second stand-in for H2, origin taken from the valley map and the geologic map over the whole network, gives the same answer: the 155,236 order-≥ 2 segments on volcanic units are shallower than the 10,799 on mapped valleys (median 2.4 against 13.9 m km^{-1}), and the two differ little on the thermal index (Cohen's $d = -0.21$; J. Cohen 1988).

5.2. *Against earlier automated mapping*

Each product here falls short of the method built for its own landform, and the comparison shows where. A convolutional network on lunar elevation models recovers 92% of a human-made test set's craters (A. Silburt et al. 2019); fill depth on the Mars model recovers 52.4% of the catalogued craters ≥ 5 km and does not see degraded or breached rims, which a learned detector was not tried on here. The global valley map of W. Luo

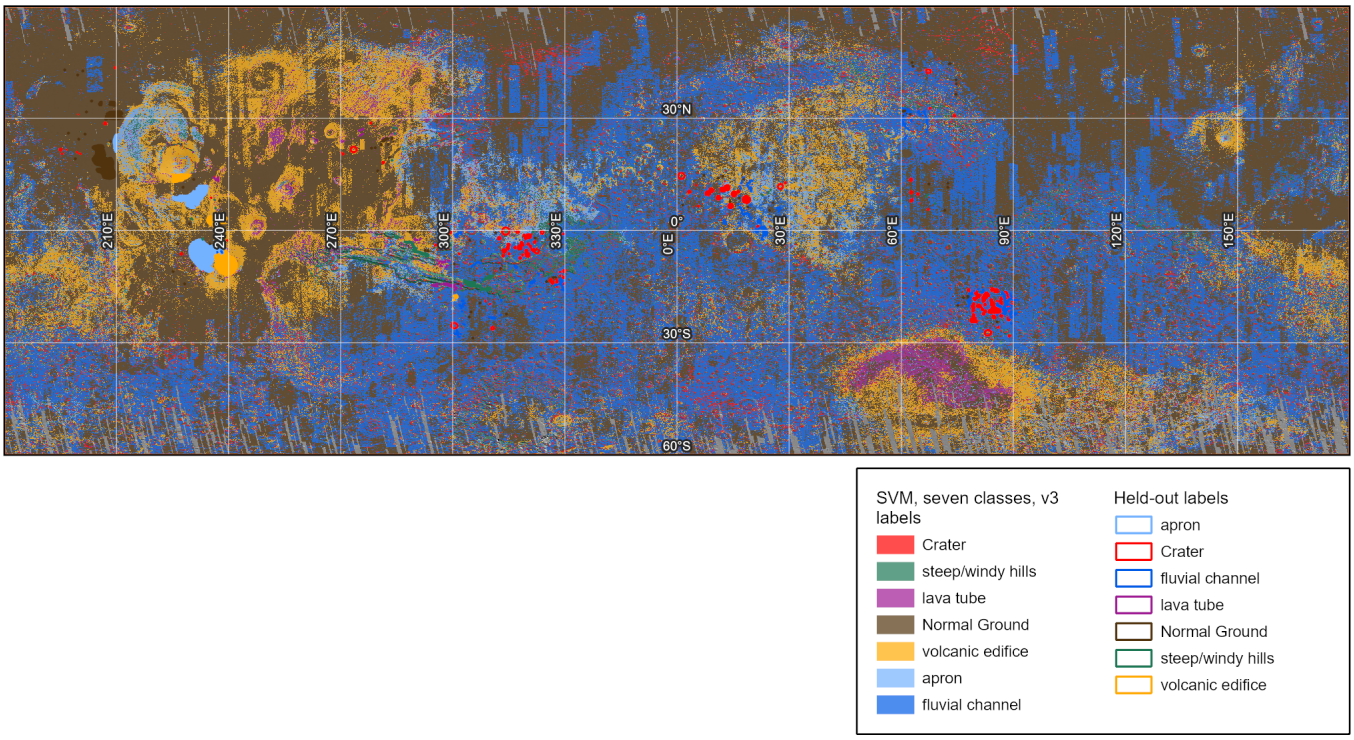

Figure 24. The support vector machine map on the third label set, with the 1,652 held-out polygons outlined in their class colours. Held out: 36.5%, κ 0.26; 39.1%, κ 0.29 on the 237 v2 polygons.

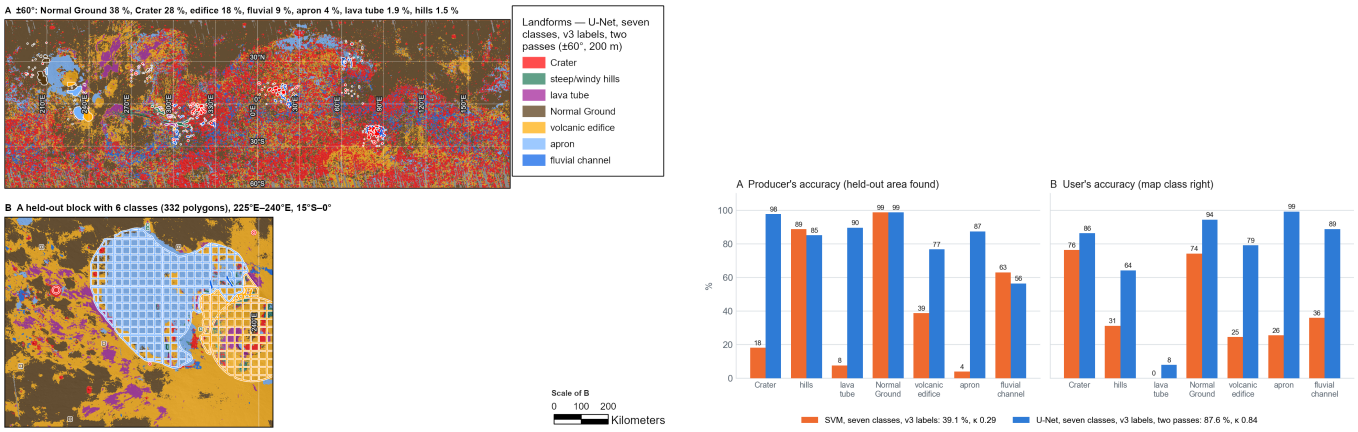

Figure 25. Left: the U-Net map on the third label set over ±60°, with a held-out block enlarged. Right: the two maps on the third label set under the same scorer, producer's (A) and user's (B) accuracy per class on the 237 v2 held-out polygons.

& T. F. Stepinski (2009) was found by an algorithm on topography and then edited by eye; unedited, the routed network here meets the valley lines of G. Alemanno et al. (2018) no more often than the same map rotated in longitude, so the editing is the method, and the candidate review stands in for it. DoMars16k (T. Wilhelm et al. 2020) trains a network on 16,150 Context Camera samples of fifteen landforms; the classes here rest on 971 hand-drawn polygons at 200 m (the third label set's 10,269 training and 1,652 held-out polygons are cut from them or placed by rule), which is why every score sits beside a geography-only baseline and why the U-Net's lead reads as a province map. Unsupervised classes from topographic attributes (B. D. Bue & T. F. Stepinski 2006) need no labels but name nothing; the supervised maps here name their classes and are scored on blocks they never saw. What the stack adds is the one grid: the three families are mapped from co-registered visible, thermal and elevation data, and the thermal bands' share is measured rather than assumed:

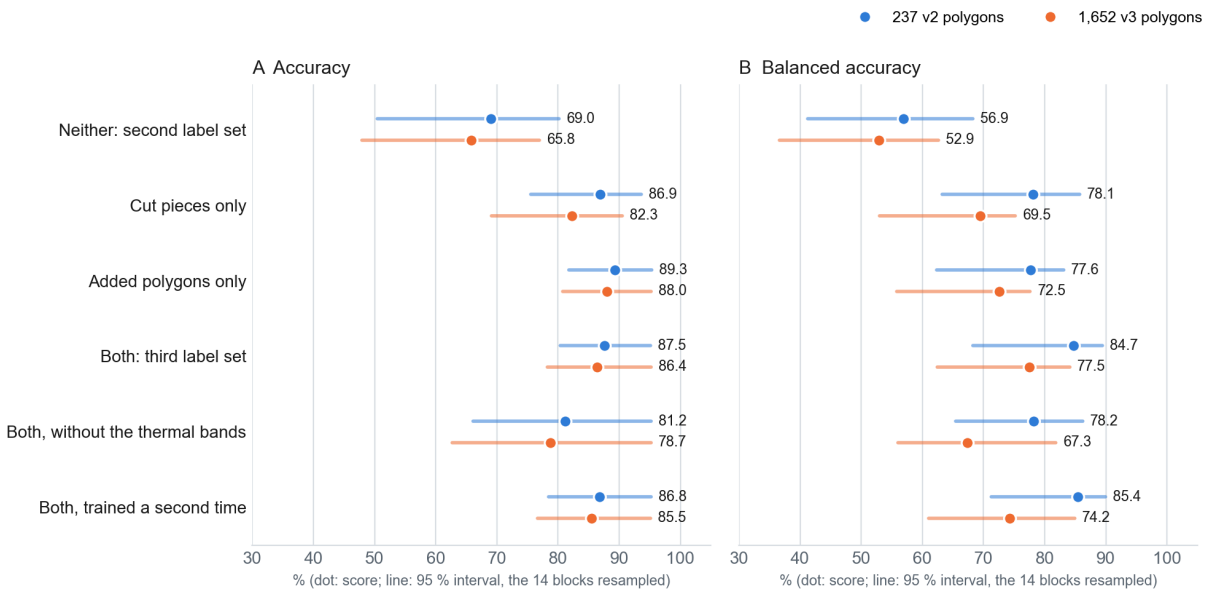

Figure 26. The U-Net by what its training labels held: each row a window map trained once with the published model's settings, scored on the 237 v2 and the 1,652 v3 held-out polygons; the first row is the published map on the second label set, the last the third label set trained a second time. Dots are scores, lines 95% intervals with the 14 blocks resampled. A: accuracy; B: balanced accuracy.

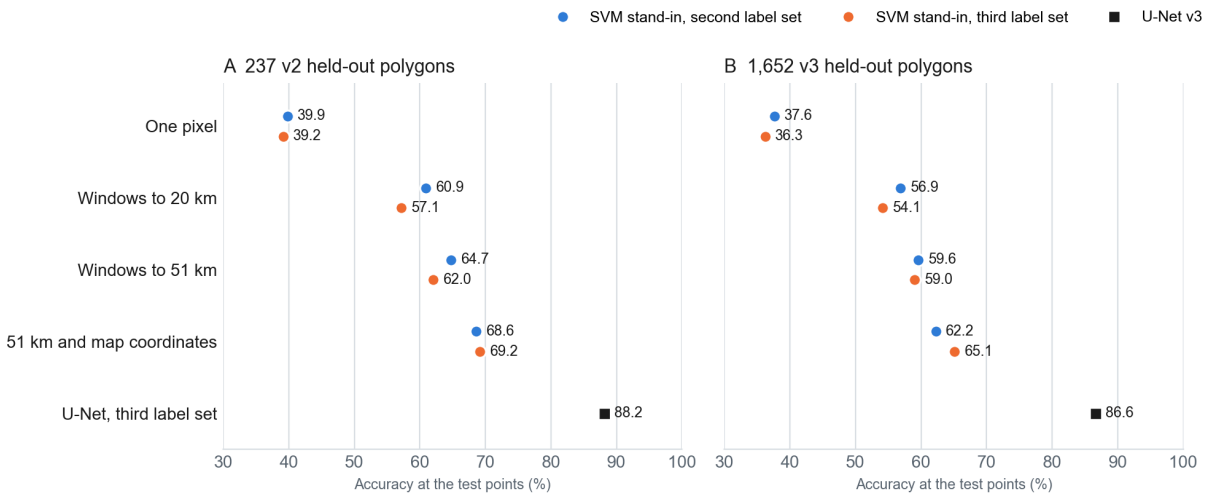

Figure 27. The support vector machine by how much ground it sees: a scikit-learn stand-in on the same stack, trained once on each label set, on one pixel, with each band's mean and spread over windows to 20 km, to 51 km, the U-Net's view, and to 51 km with the point's map coordinates; the published U-Net map read at the same 14,993 (A) and 14,996 (B) test points.

for the support vector machine small (1.4 to 2.0 points on four classes) and, by balanced accuracy, absent; for the seven-class U-Net, larger (6.4 to 7.6 points, where a second training with them moves under one) but not resolved by block; and for a support vector machine given the U-Net's 51 km of surroundings, 8.0 to 9.9 points and resolved, all of it in the surroundings and most or all of it recovered by the point's map coordinates instead, so it is not shown to be material.

5.3. *Limits*

The central limit is in the data held. The THEMIS mosaics used here are display products, stretched region by region, so they carry local contrast and texture; planet-wide material discrimination needs calibrated products such as the TES inertia map, at the cost of resolution (3 km), or THEMIS-derived inertia (R. L. Fergason et al. 2006) where it exists. In this study the

Table 5. The hypotheses and tests

	Claim	Verdict	Decisive number
H1	Day IR shows flow margins Viking misses	not supported, both tests	day − Viking +4.9 pt, interval −2.7 to +13.1
H2	Fluvial and volcanic channels separate	not supported (stand-ins)	volcanic shallower; thermal $\|d\| \leq 0.31$
H3	Composite classes are most stable	not supported	ARI 0.40 against 0.61–0.89
H4	Large IAU craters recovered	answered by fill depth	68% of 117 craters ≥ 100 km
T1	Thermal IR improves the classification	slightly, by accuracy only	+1.36 and +1.97 pt, intervals above 0
T3	THEMIS tracks calibrated inertia	weakly, locally	within-block $\|r\| \approx 0.1$; Viking −0.36
T7	Tributary orders resolved at 100 m	no DEM-supported answer	order follows threshold

Viking colour carries the regional material signal, and it is not calibrated albedo.

The labels limit what the classification can show. The four hand-drawn classes are landforms, and slope and relief alone predict them as well as the full stack; a test of thermal value needs material labels, which only photo interpretation can give. The geologic map's unit groups (K. L. Tanaka et al. 2014) are no substitute: on a 4 km lattice of the stack they are predicted no better than by their commonest group, with or without the thermal bands, which add at most a point. Lava tube, with 44 polygons, is too thin a class for a planet-scale map; the U-Net makes it usable only as a province.

An audit of the 512 labels found no two classes sharing ground, but five Crater polygons drawn around volcanoes (Arsia, Pavonis and Ascraeus Mons, Tharsis and Ulysses Tholus) and six lava-tube polygons beside fluvial valleys. Arsia Mons alone is 31% of the held-out Crater area, so part of the Crater score, and of the U-Net's Crater province, measures volcanic shields. A seven-class scheme follows from the audit: Crater narrowed to impact craters, with volcanic edifice, apron and fluvial channel as new classes. Its 971 labels (734 training, 237 held out in the same 14 blocks) were applied on 27 October. Every relabel, check and new edifice outline was reviewed by eye; the 172 apron and 276 fluvial-strip proposals were screened by a rule written before it ran (a strip kept only where its median height lies at least 20 m below that of a 4.8 km flank ring and under half of it lies on volcanic plains; an apron only where its median slope is at least 1° and under half of it lies inside crater rims), not inspected one by one; and hills fall short of the training minimum, in two training blocks against three. Both seven-class maps failed the rule set before they ran, on the four original classes, and are presented for what they name rather than for that score: the support vector machine over-predicts fluvial channel and the U-Net volcanic edifice. On the third label set the U-Net's edifice falls to 18.0% of ±60°, while the support vector machine's fluvial channel stays at 31.9%. Off Tharsis the edifice class is weak: on six polygons the AI assistant drew at the calderas of Nili and Meroe Paterae on Syrtis Major, in a held-out block, the U-Net calls 7.7% of the ground volcanic edifice (1.4–20.5%, resampling the polygons) and the support vector machine 0.6%, so the U-Net's edifice score is Tharsis ground; the assistant could not tell lobate debris aprons from talus on the views it had, so apron was not scored there.

The detectors are elevation-model products. Closed depressions miss breached craters by construction, and below about 5 km their density follows the sharpness of the HRSC/MOLA blend. Channels inherit the same dependence, and the fill mask that removes synthetic lakes also breaks networks where the DEM is pitted.

Every polygon score in this paper is taken where a class is clear, which flatters five of the six maps sampled, all but the support vector machine on the third label set. The probability sample shows the size of that, but on the AI assistant's labels, which failed their check; unbiased accuracies and class areas need the same points labelled by an analyst who passes it. The U-Net maps are firmer as scores than as pixels: the third label set's model, run again over the held-out windows on chips placed differently, scores 87.54% against 87.62% on the 237 polygons, yet about one pixel in ten changes class, so a single pixel of a U-Net map should not be read alone. Nor do the polygon scores fix the class areas. Trained a second time with every setting the same, the U-Net scores within a point of the first on both polygon sets, but over the held-out windows its volcanic edifice covers 8.7% of the ground against 16.3%, the difference going mostly to Crater and Normal Ground. At the 39 labelled sample points inside the windows the second training's map calls 4 volcanic edifice against 10, nearer the AI assistant's 1, too few points to choose between them. The edifice share of ±60° (18.0%) is one training's.

The held-out blocks have now been used three times, by the four-class models, the seven-class ones and those on the third label set. That set was requested after the seven-class scores were read, so any gain on it is not a fresh test, and its held-out crater discs come from the catalogue the crater results are checked against, so that

class partly measures agreement with it. Fourteen other blocks, drawn at random, give a retrained U-Net 86.6%, and a second draw 85.4%, so its score is not tied to the first 14, though on the second its lead over the nearest training polygon is resolved only once weighted to the first 14 blocks' class mix; the four- and seven-class scores still rest on the first 14. Any further tuning, of the deep-learning models above all, would turn the 14 held-out blocks into a validation set; a next round needs validation blocks of its own. Finally, the manual review of the 9,287 type-area candidates (2,610 and 3,283 channels, 1,685 and 1,709 depressions) has reached 3,791, with a precision (accepts over accepts and rejects) of 54% on the 188 steep, rock-floored Ius channels, 50% on 209 rock-floored Athabasca channels, and 10% and 13% on the closed depressions of the two areas. The AI assistant read the other 200 rock-floored Athabasca channels on CTX, at the author's decision, and accepted 3 (186 rejected, 11 unsure); on 48 lines the author had reviewed, the same reading accepted 3 against the author's 22 and agreed with the author no better than chance ($\kappa = 0.14$), so the channel precisions are each reader's standard, not ground truth. The digitised classes hold 202 channels (199 accepted by the author, 3 by the assistant) and 381 crater rims, and the 86 traced lava margins, so the machine products are presented as candidates with measured recall and precision.

5.4. *Future work*

Five steps would move these results furthest. The probability sample needs labels from an analyst who passes its check (Section 4.10), to give unbiased accuracies and class areas with intervals, free of where the polygons were drawn. Material labels from photo interpretation, rather than landforms or the geologic map's unit groups, would test what the thermal bands add, and THEMIS-derived inertia (R. L. Fergason et al. 2006), where it exists, would replace the display mosaics' local contrast. A check of the lava margins traced on Context Camera images (M. C. Malin et al. 2007) by a reader other than the assistant that traced them would make H1's test independent; H2 needs a study area that holds both fluvial and volcanic channels. The deep-learning models need validation blocks of their own before any retraining, and labels checked by eye rather than placed by rule: the lead U-Net's lava tube (user's accuracy 8.0%) and its volcanic edifice, denser inside none of ten small named edifices, come first. For the craters, a rim-based diameter or merged nested basins would recover the 23% of missed craters that hold only a smaller closed basin, 72% of the misses at ≥ 50 km.

6. CONCLUSIONS

1. Four global products now share one co-registered grid over $\pm 60°$, 86.6% of Mars, with the THEMIS pair never resampled and the elevation model reproduced exactly at 200 m.

2. Viking colour, day infrared and night infrared span about three dimensions locally, night infrared nearly independent of the others, but the held THEMIS mosaics carry local contrast only; calibrated thermal inertia follows Viking colour across the planet.

3. Fill depth recovers 68% of the large named craters and gives 465,354 closed depressions ≥ 1 km whose precision against Robbins rises from 15.3% to 63.2% at 5 km; recall, 52.4% at 5 km, falls with rim degradation, from 83.1% for the freshest rims to 38.4% for the most degraded, and breached craters stay invisible.

4. Routed channels are dominated by the elevation model at $\pm 60°$ (89.8% under 2°) and meet the mapped valleys no more often than a rotated valley map; stream order is a parameter, not a measurement.

5. The four-class landform map scores a balanced accuracy of 66.7% (73.8%, κ 0.58 overall) on blocks it never saw, where geography alone, the nearest training polygon's class, gives 61.3% (83.2% overall); the thermal bands add a resolved 1.4 to 2.0 points of overall accuracy and none by balanced accuracy. A seven-class U-Net trained once without them scores 6.4 to 7.6 points lower, a difference not resolved by block but larger than a second training's under one point; a support vector machine given 51 km of surroundings loses 8.0 points without them, a resolved difference that the point's map coordinates largely make up. A U-Net reaches 98.9% as a terrain-province map.

6. Revised to seven classes, the maps separate impact craters from volcanoes and name volcanic edifices, aprons and fluvial channels: the U-Net scored 69.0%, κ 0.606 and the support vector machine 42.5% on 237 held-out polygons, below the four-class maps on the original classes.

7. On a third label set of more and smaller polygons the U-Net scores 87.6%, κ 0.844 on the same polygons and 49.6% against 21.7% on a probability sample, a resolved gain on both, though the sample's labels failed their check and the set was

asked for after the earlier scores were read; on two draws of 14 other blocks, held out from retrained models, it scores 86.6% and 85.4%, and its gain over the second label set comes about equally from cutting the polygons and from adding new ones. The support vector machine gains nothing resolved (39.1%, κ 0.288; 40.2% on the sample) and fails the rule set for it, held by one pixel's seven bands rather than by its sample of 1,000 pixels a class or by its labels; both lead this paper.

8. Lava flows are not separable in these bands; the geologic map remains the lava-flow reference; H1 is not supported on 1,476 km of margins traced on CTX either, and H2 awaits a window that holds both fluvial and volcanic channels.

DATA AVAILABILITY

The ±60° landform maps, the type-area products, the candidate and label layers, the trained models, every build script and the project record are published at L. M. Edwards (2026): the scripts, the record and the vector layers in the repository, the rasters and models as the files of its release `data-2026-11-10`; the code under the MIT licence and everything else under CC BY 4.0. A script there regenerates every product and figure, and the repository lists the software and versions that each machine ran. The source mosaics, the elevation model, the geologic map, the thermal-inertia map and the crater database are public from the sources cited in Section 2.

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Facilities: Mars Odyssey (THEMIS), Viking Orbiter, MGS (MOLA, TES), Mars Express (HRSC)

Software: ArcGIS Pro (Esri 2026), GDAL (GDAL/OGR contributors 2024), NumPy (C. R. Harris et al. 2020), Matplotlib (J. D. Hunter 2007), PyTorch (A. Paszke et al. 2019)

APPENDIX

A. THE RULES OF THE THIRD LABEL SET

The third label set (Section 4.9) holds 11,921 polygons, 10,269 for training and 1,652 held out. No one accepted or rejected a polygon by eye: each row was passed or failed by a rule written into the script before it ran, and the rules are given here in full so that the set can be rebuilt or changed. Two kinds of polygon enter it. A *piece* is part of a polygon of the second label set: every such polygon over 1,500 km^2 or wider than 1° is cut by a 0.5° grid, pieces under 25 km^2 are dropped, and a piece is kept if it passes the screen of its class; smaller polygons are kept whole and unscreened. A piece keeps its parent's side of the split, so no held-out ground of the second set becomes training ground. A *new polygon* is placed from reference data alone, never on the ground of any polygon of the second set, kept or dropped, and one cell clear of every other new polygon; it goes to training or held out by the 15° block of its centre, and its number is capped in each block.

The measures come from three sources. The 200 m elevation model is summarised in cells of 16×16 pixels (3.2 km): the median and the 90th-percentile slope, and the 5th, 50th and 95th-percentile height. The crater database (S. J. Robbins & B. M. Hynek 2012) is drawn as discs on an 800 m grid at 1.0 crater radius, the rim test, and at 1.3 radii for craters of 3–100 km, the exclusion; "crater cover" is the share of a polygon inside those discs. The geologic map (K. L. Tanaka et al. 2014) gives each cell a unit kind: edifice, apron, landslide, volcanic plains or highland. Table 6 counts the polygons each class gave; the screen of a cut piece and the rule that places a new polygon, with its cap in a 15° block, follow class by class.

1. *Crater.* A piece is kept if at least 0.5 of it lies inside the rims of catalogued craters ≥ 4 km. A new polygon is a catalogued crater's disc at the rim, one in each diameter bin (6–12, 12–25, 25–50 and 50–120 km) in a block, placed if the rim stands at least 50 m, or 5 m per km of diameter if that is more, above the floor.

Table 6. The third label set: polygons for training and held out, with the 15° blocks that hold them, as first written; the fan-shaped deposits then raised apron to 1,924 and 356.

Class	Training (blocks)	Held out (blocks)
Crater	949 (178)	305 (14)
Steep/windy hills	631 (150)	121 (11)
Lava tube	106 (11)	51 (5)
Normal Ground	2,081 (178)	319 (14)
Volcanic edifice	1,637 (23)	207 (3)
Apron	1,868 (21)	306 (3)
Fluvial channel	2,941 (111)	293 (8)

2. *Steep/windy hills.* A piece is kept if the median of its cells' 90th-percentile slope is $\geq 8°$ and under half of it lies on edifice, apron or landslide units. A new polygon is 3×3 cells with at least 7 of 9 at a 90th-percentile slope $\geq 10°$, relief $\geq 1{,}000$ m and no crater ≥ 10 km within 1.3 radii, off those units; 4 a block.

3. *Lava tube.* Polygons are kept whole, since no data screens a piece of a pit chain. New polygons are troughs in the mapped extent of the 16 named catenae: a black top-hat of the elevation model (4.2 km closing) at least 50 m deep, with an elongation ≥ 3 and a length ≥ 6 km, outside crater rims; at most 15 a catena.

4. *Normal Ground.* A piece is kept if its median slope is $\leq 2°$, its crater cover ≤ 0.3 and it lies off those units; craters of 8–100 km are then cut out at 1.2 radii, and half the piece must remain. A new polygon is 4×4 cells, every cell at a median slope $\leq 1°$, a 90th-percentile slope $\leq 3°$ and relief ≤ 150 m, with no crater ≥ 4 km within 1.3 radii, off those units; 5 a block.

5. *Volcanic edifice.* A piece is kept if it lies on edifice units or a named volcano and slopes $\geq 1°$, unless it is within 0.3 of the volcano's radius, where calderas are flat. A new polygon is 4×4 cells of edifice units, every cell sloping $\geq 1°$, with no crater ≥ 10 km within 1.3 radii; 12 a block.

6. *Apron.* A piece is kept if it lies on apron units and slopes $\geq 0.7°$; poleward of 30° its cells' 90th-percentile slope must also be $\leq 15°$, which keeps mesa scarps out. A new polygon is 4×4 cells, at least 14 of 16 passing the same tests; 12 a block.

7. *Fluvial channel.* A piece is kept if its floor lies at least 20 m below a flank ring 4.8 km out and under half of it lies on volcanic units. New polygons are strips 20 km long and 2.4 km wide along every second segment of the mapped valleys (G. Alemanno et al. 2018), under the same floor rule; 2 a valley and 6 a block.

The screens rejected 161 Crater pieces that lay outside rims, 345 edifice, 146 apron, 47 Normal Ground and 910 fluvial pieces, and 680 new valley strips. Of 1,049 craters measured for a new disc 693 were placed, of 1,269 valley strips 589, and of 592 troughs 119. Against the second label set the hills train in 150 blocks where they trained in 2, and the lava tubes in 11 where they trained in 3; the edifice polygons still train in 23 blocks and are held out in 3, which is why their score is Tharsis ground (Section 4.9).

Two rules were changed after the first polygons were read on hillshade sheets and before the set that stands was written. The first run placed 315 lava-tube chains from closed depressions outside crater rims; at full resolution none showed a pit, so the catena troughs replaced them. The cap on the cells' 90th-percentile slope had rejected the ridged aureole lobes of Olympus Mons, which the class key names as apron, so it now applies poleward of 30° only. One rule was changed after a check of the screened rows: the slope test rejected the flat fan-shaped deposits on the west flanks of the Tharsis volcanoes, so those deposits pass without it, which added 106 apron polygons, all mostly on that unit.

B. PREDICTIONS WRITTEN BEFORE EACH RUN, AND THEIR OUTCOMES

Each run behind the lead maps had its predictions, and where a choice hung on it a rule, written into its script before it ran; nothing was adjusted afterwards, and each prediction is reported held or not held whichever way it fell. They are listed here in the order the runs were made, with the number that decided each. Of the 28 below, 20 held. A prediction that failed is as much a result as one that held: the U-Net's score on the third label set was above the range written for it, which is why a leak was sought before the score was believed (Section 4.9).

The maps on the third label set (nine, written while the U-Net trained and before any map existed; five held). The 237 polygons are the second label set's held-out polygons, the 1,652 the third's.

P1 The U-Net's balanced accuracy on the 237 polygons is above the earlier U-Net's 56.9%. *Held*: 84.5%.

P2 Its accuracy there lies in 60–78%. *Not held*: 87.6%, above the range.

P3 The support vector machine's accuracy there lies in 37.5–50%. *Held*: 39.1%.

P4 The hills' producer's accuracy rises on both models. *Held*: 50.3 to 88.7% and 46.8 to 85.1%.

P5 The U-Net maps volcanic edifice on under 40.0% of ±60°. *Held*: 18.0%.

P6 The U-Net's gain over the earlier U-Net is not resolved when the blocks are resampled. *Not held*: the intervals of accuracy and of balanced accuracy both lie above 0.

P7 The support vector machine does not separate from the nearest training polygon on the 1,652 polygons. *Held*: −21.7 to +20.3 points of accuracy.

P8 The U-Net's lead over the support vector machine by balanced accuracy is resolved on the 237 polygons and not on the 1,652. *Not held*: it is resolved on both, +18.4 to +47.1 and +16.9 to +40.4 points.

P9 On the 218 polygons new to the held-out set both models beat their earlier maps. *Not held*: the U-Net rises from 36.1 to 81.1%, the support vector machine falls from 30.3 to 25.9%.

Fourteen other blocks held out, twice (ten; nine held). The models were retrained without 14 blocks drawn at random from the training blocks, and scored on them.

F1 First draw: the U-Net's accuracy lies in 70–92%. *Held*: 86.6%.

F2 Its balanced accuracy lies in 60–85%. *Held*: 72.2%.

F3 The U-Net leads the support vector machine, resolved by block. *Held*.

F4 The support vector machine does not separate from the nearest training polygon. *Held*.

F5 The U-Net leads the nearest training polygon, resolved by block. *Held*.

F6 The second draw repeats all five. *Four held*: 85.4% and 69.2%, the lead over the support vector machine resolved, the support vector machine not separated from the nearest polygon. *The fifth did not*: the lead over the nearest training polygon is not resolved by accuracy on that draw, whose polygons lean to apron.

The U-Net without the thermal bands (two; one held).

T1 Its accuracy is within 3 points of the seven-band model's on each polygon set. *Not held*: 6.4 and 7.6 points lower.

T2 The difference is not resolved when the blocks are resampled. *Held* on each set.

Cutting the polygons and adding new ones, apart (four; three held).

C1 Trained on the cut pieces alone, the U-Net's balanced accuracy on the 237 polygons is above 56.9%. *Held*: 78.1%.

C2 Trained on the added polygons with the uncut second label set, it is above 56.9% too. *Held*: 77.6%.

C3 Adding gives more than cutting. *Not held*: 77.6% against 78.1%, a difference not resolved either way.

C4 Each stays below the two together. *Held*: both are below 84.7%.

A second training with the same labels and settings (three; two held).

R1 Its accuracy is within 3 points of the first training's on each polygon set. *Held*: 0.74 and 0.87 points apart.

R2 Its map differs from the first training's on less of the windows' ground than the map without the thermal bands does. *Held*, narrowly: 26.8% against 28.1%.

R3 Its validation loss rises above 5 late in training, as the first training's did. *Not held*: it peaks at 2.09 where the first reached 23.25, so the late rises belong to a training and not to the labels.

Table 7. Producer's accuracy by class, and the overall scores, of each U-Net training (per cent; κ as a fraction).

	second set	third set	trained again	no thermal	cut only	added only
On the 237 polygons						
Crater	57.8	97.4	98.4	98.3	80.4	97.1
steep/windy hills	46.8	86.6	97.7	62.1	76.6	49.2
lava tube	22.4	90.5	95.6	83.1	44.8	51.9
Normal Ground	69.7	98.6	98.8	98.9	97.5	98.0
volcanic edifice	81.3	76.3	66.6	87.1	94.6	86.4
apron	77.4	87.4	90.4	61.1	85.4	89.2
fluvial channel	43.0	56.4	50.0	56.4	67.6	71.9
Accuracy	69.0	87.5	86.8	81.2	86.9	89.3
Balanced accuracy	56.9	84.7	85.4	78.2	78.1	77.6
κ	0.606	0.843	0.833	0.763	0.834	0.864
On the 1,652 polygons						
Crater	52.5	94.0	92.8	94.2	72.5	95.8
steep/windy hills	42.4	80.6	92.3	59.0	68.6	44.2
lava tube	4.9	46.7	27.4	15.0	7.6	26.4
Normal Ground	66.9	97.7	98.0	98.0	93.6	96.0
volcanic edifice	82.8	79.1	69.1	89.7	96.8	88.8
apron	73.8	82.0	85.4	54.7	81.5	85.6
fluvial channel	47.0	62.4	54.8	60.6	65.6	71.0
Accuracy	65.8	86.4	85.5	78.7	82.3	88.0
Balanced accuracy	52.9	77.5	74.2	67.3	69.5	72.5
κ	0.565	0.827	0.815	0.731	0.774	0.845

Table 8. Accuracy on the 237 polygons in each of the 14 held-out 15° blocks (per cent). A block is named by its south-west corner in units of 15°, east longitude then latitude.

Block	second set	third set	trained again	no thermal	cut only	added only
−11, 1	77.7	99.9	100.0	100.0	99.2	99.7
−11, 2	54.3	100.0	100.0	100.0	99.4	99.5
−3, −1	73.7	93.8	89.3	93.5	91.7	92.8
−4, −1	39.2	84.9	90.8	79.4	76.0	60.4
−4, −2	31.8	45.4	53.8	36.0	44.6	38.1
−5, −1	89.7	96.7	98.9	52.6	95.2	56.7
−6, 1	10.6	56.7	78.7	79.8	50.7	47.7
−9, −1	68.6	77.0	74.0	56.2	84.6	83.4
−9, 0	96.3	94.7	93.9	96.9	97.8	97.5
0, 0	13.9	98.3	98.4	99.2	36.5	97.9
1, 0	14.7	77.5	72.8	72.4	58.7	85.9
4, 0	66.7	88.2	99.7	94.0	99.9	94.2
4, 2	11.9	47.2	20.2	35.0	27.8	22.7
5, −2	74.2	89.1	92.7	93.2	91.4	95.3

C. THE U-NET TRAININGS CLASS BY CLASS AND BLOCK BY BLOCK

Tables 7 and 8 give what the text summarises: every U-Net training on the seven classes, scored on the same windows by the same scorer. "Second set" is the U-Net on the second label set; "third set" the lead model's labels and settings classified on the windows in one pass, which is why its figures differ by a few tenths from the lead map's; "trained

again" the same labels and settings a second time; "no thermal" the same labels on five bands; "cut only" and "added only" the two halves of the third label set's change (Section 4.9). Each column is one training.

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